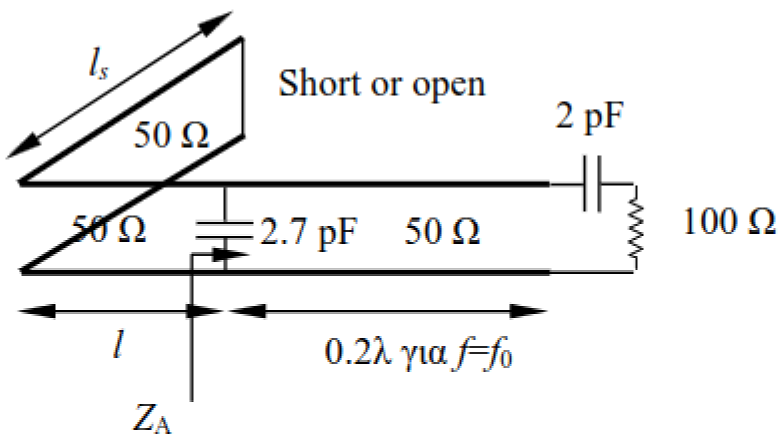
TRANSMISSION LINES



1.1 Study of a transmission line circuit as a function of frequency - Matching

The following TEM transmission line circuit is given with a characteristic impedance of 50 Ω operating at a frequency f_0 = 1 GHz (let λ be the wavelength in the transmission line for this frequency).

(a) Calculate, using the Smith chart, the complex input impedance Z_A, exactly to the left of the parallel capacitor, as shown in the figure

Solution:
To calculate the complex input impedance Z_A using the Smith chart, we follow the steps below, starting from the load and proceeding toward the generator.

1. Load Normalization

The load consists of a 100 Ω resistor in series with a 2 pF capacitor. At frequency f_0=1 GHz, the reactance of the capacitor is:

$$X_C = -\frac{1}{2\pi f_0 C} = -\frac{1}{2\pi \cdot 10^9 \cdot 2 \cdot 10^{-12}} \approx -79.58\Omega$$

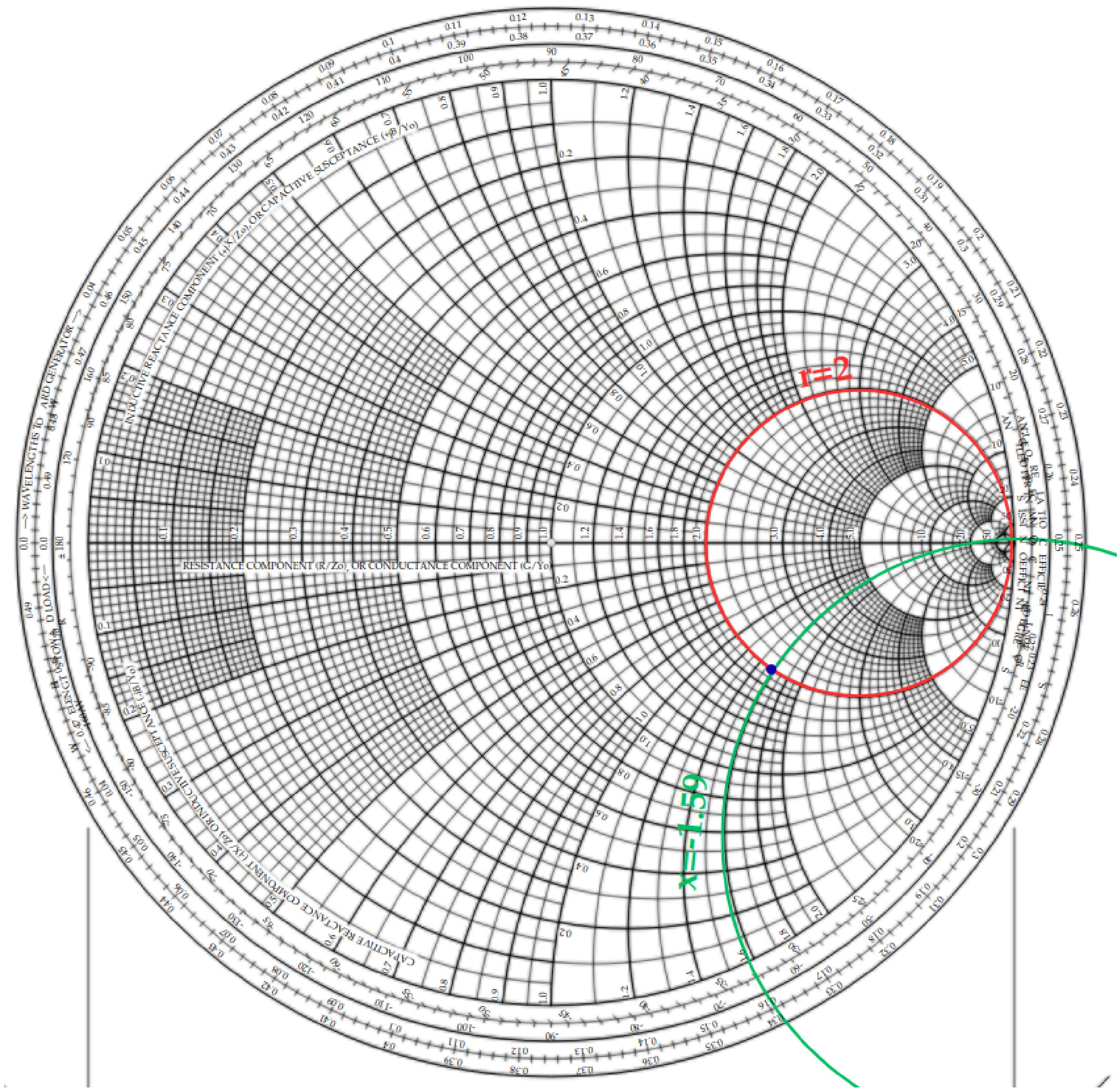
Therefore, the total complex load impedance is Z_L = 100–j79.58 Ω. Normalizing with respect to Z_0 = 50 Ω:

$$z_L = \frac{Z_L}{50} = 2 - j1.59$$

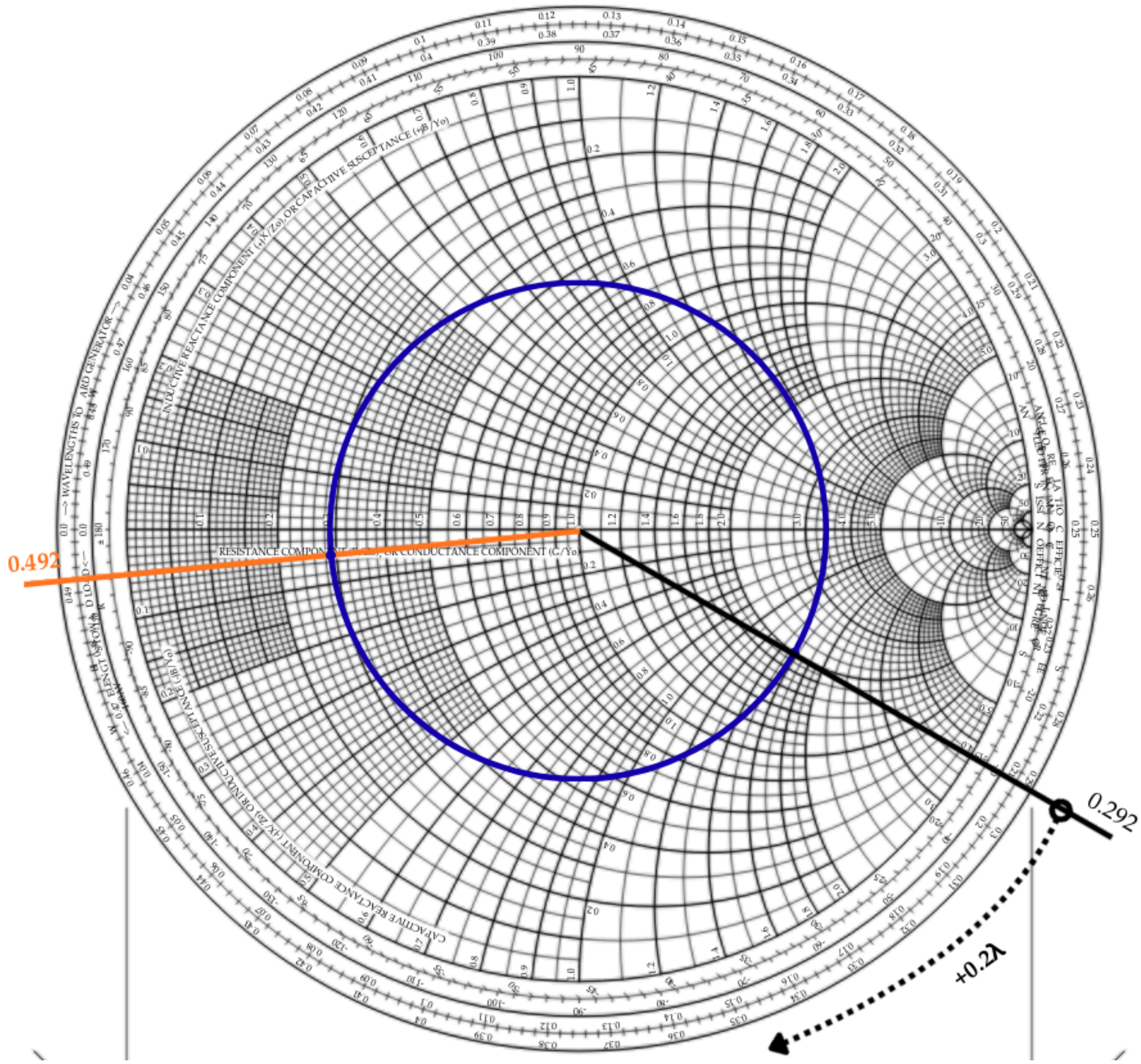
2. Movement along the Transmission Line

We must move from the load toward the generator over a length of 0.2λ.

1. We locate z_L on the Smith chart:



2. We rotate by 0.2λ clockwise (toward the generator) on the constant SWR circle.

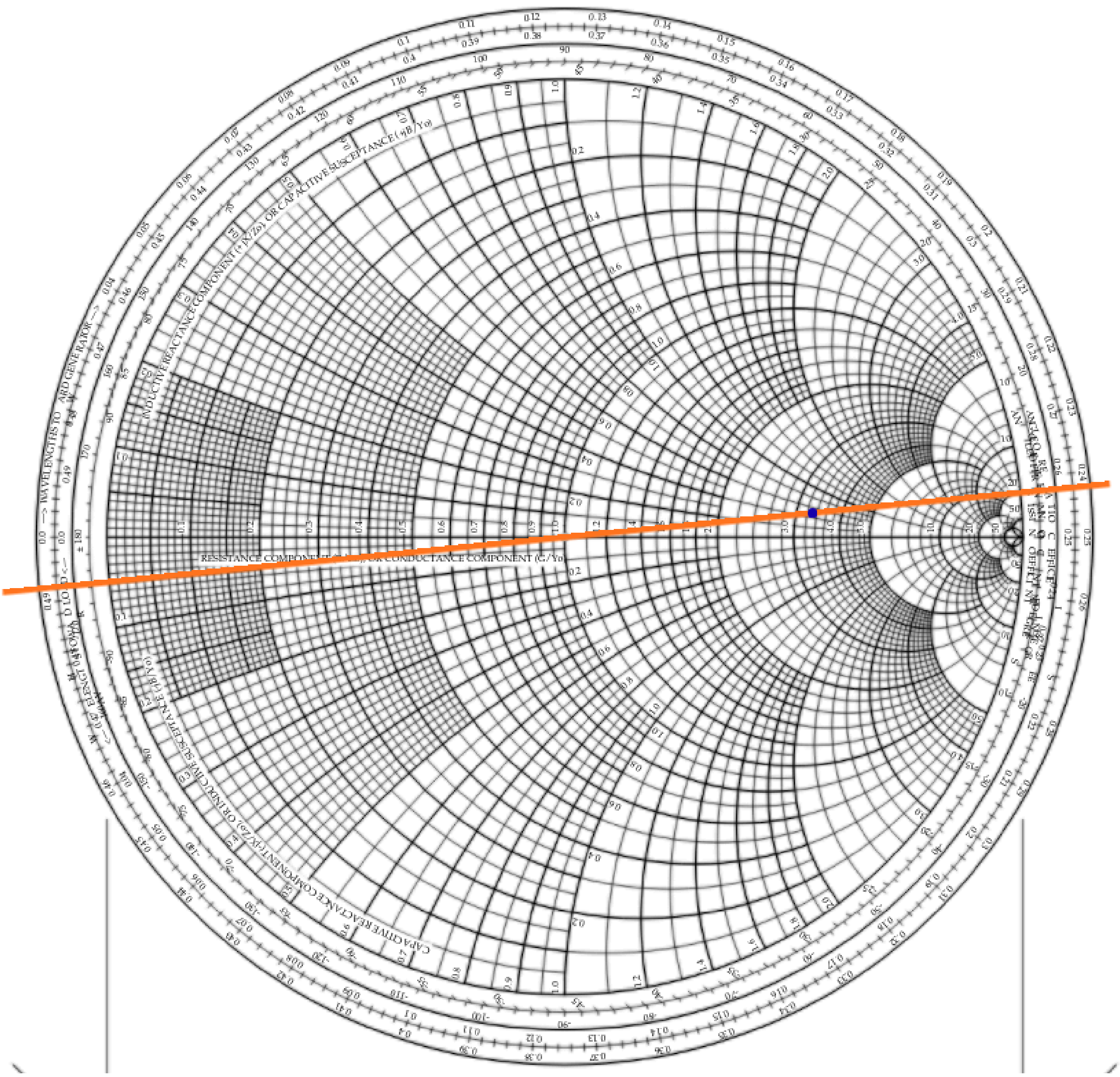


3. The resulting point is the normalized complex impedance z_{line} exactly before the parallel capacitor. From the chart, this value is approximately:
z_{line} ≈ 0.3−j0.05 (since it belongs to the circles with r=0.3 and x=-0.05)

3. Parallel Capacitor (Admittance Domain)

Since the 2.7 pF capacitor is in parallel, it is easier to work with complex admittances (Y).

1. Conversion to Admittance: We find $y_{line} = \dfrac{1}{z_{line}}$ on the Smith chart (point diametrically opposite to the center).
   $y_{line} \approx$ 3.3+j0.55 (since it belongs to the circles with r=3.3 and x=0.55)

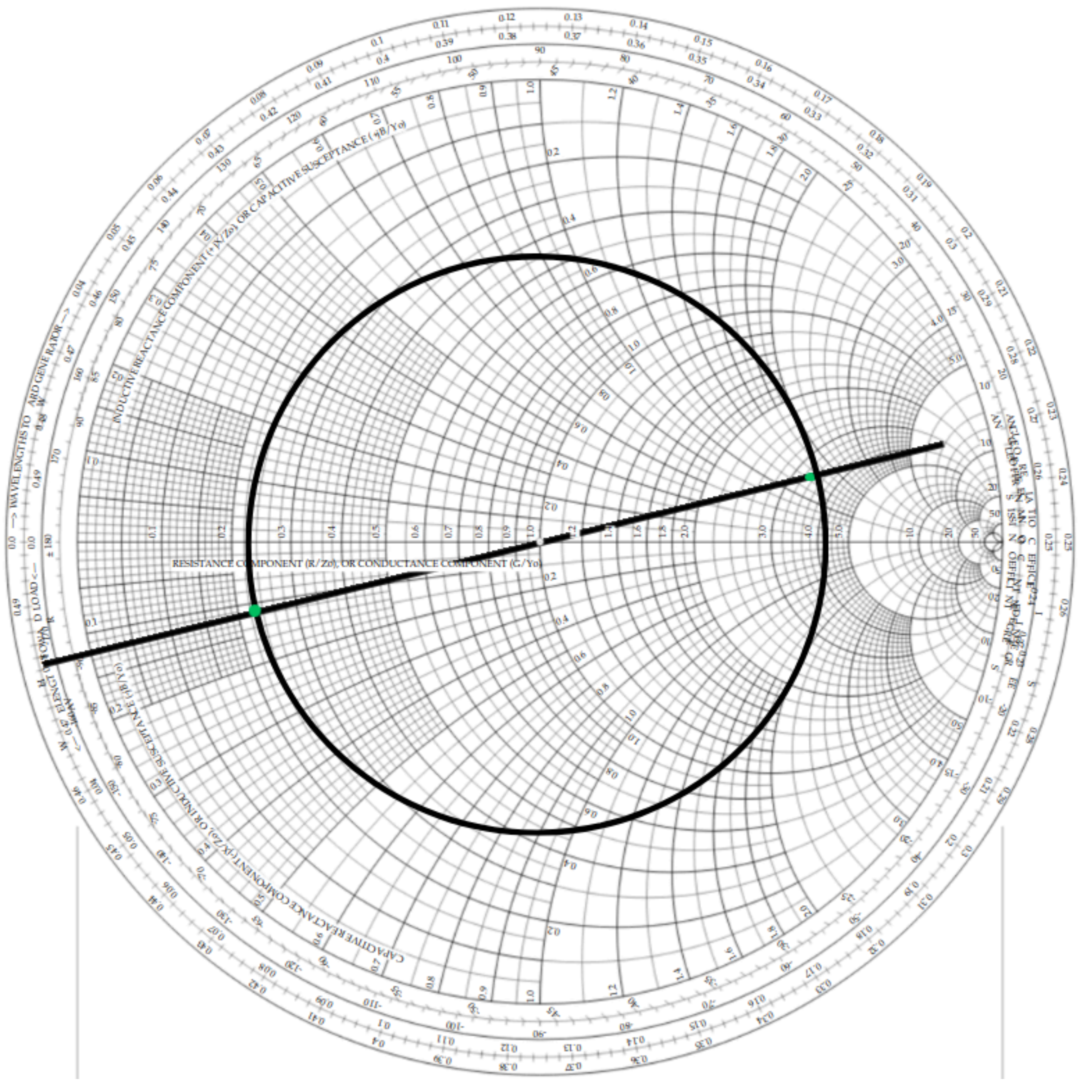


2. Normalized Susceptance: The susceptance of the 2.7 pF capacitor at 1 GHz is:
   $B_C = 2\pi f_0 C_p = 2\pi \cdot 10^9 \cdot 2.7 \cdot 10^{-12} \approx 0.017\ \mathrm{S}$
   The normalized value is: $b_C = B_C \cdot Z_0 = 0.017 \cdot 50 \approx 0.85$
3. Adding Susceptance: We move along the constant conductance circle by adding +j0.85:

   $y_A = y_{line} + jb_C = 3.3 + j(0.55 + 0.85) = 3.3 + j1.4$

4. Final Complex Impedance ZA

1. Return to Impedance: We convert y_A back to normalized impedance z_A on the chart: z_A ≈ 1/(3.3+j1.4) ≈ 0.23−j0.11



2. Denormalization: $Z_A = z_A \cdot 50$ ≈ 11.5−j5.5 Ω

(a) Solution/Verification with tool: https://onlinesmithchart.com

Link:
https://onlinesmithchart.com/?zo=1&frequencyUnit=GHz&frequency=1&zMarkers=2_-1.59&circuit=blackBox_2_-1.59_0__transmissionLine_0.2_%CE%BB_0_1_1__shortedCap_135_pF_0_0_0&vswrCircles=3.48

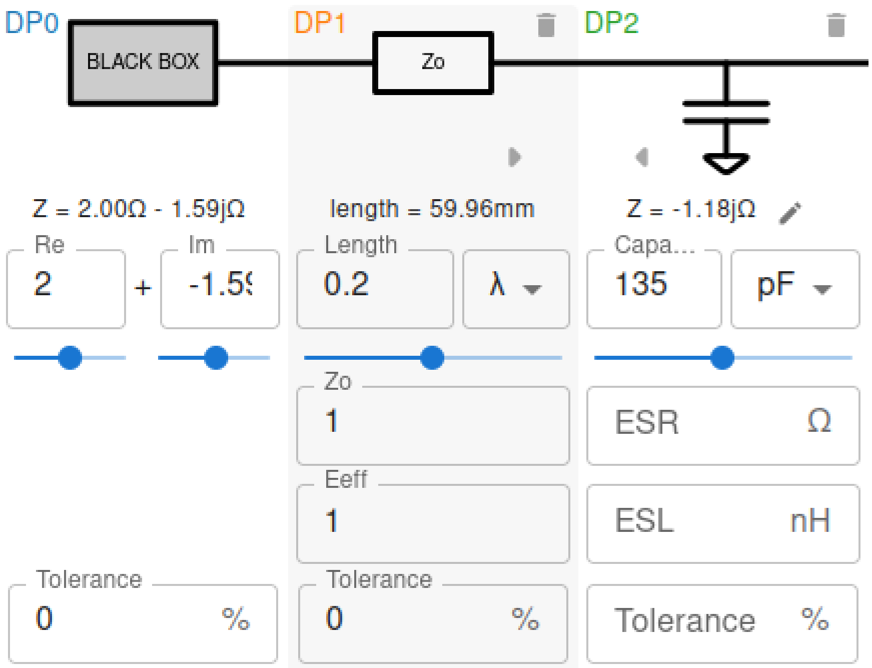
1. Load Normalization

(...)

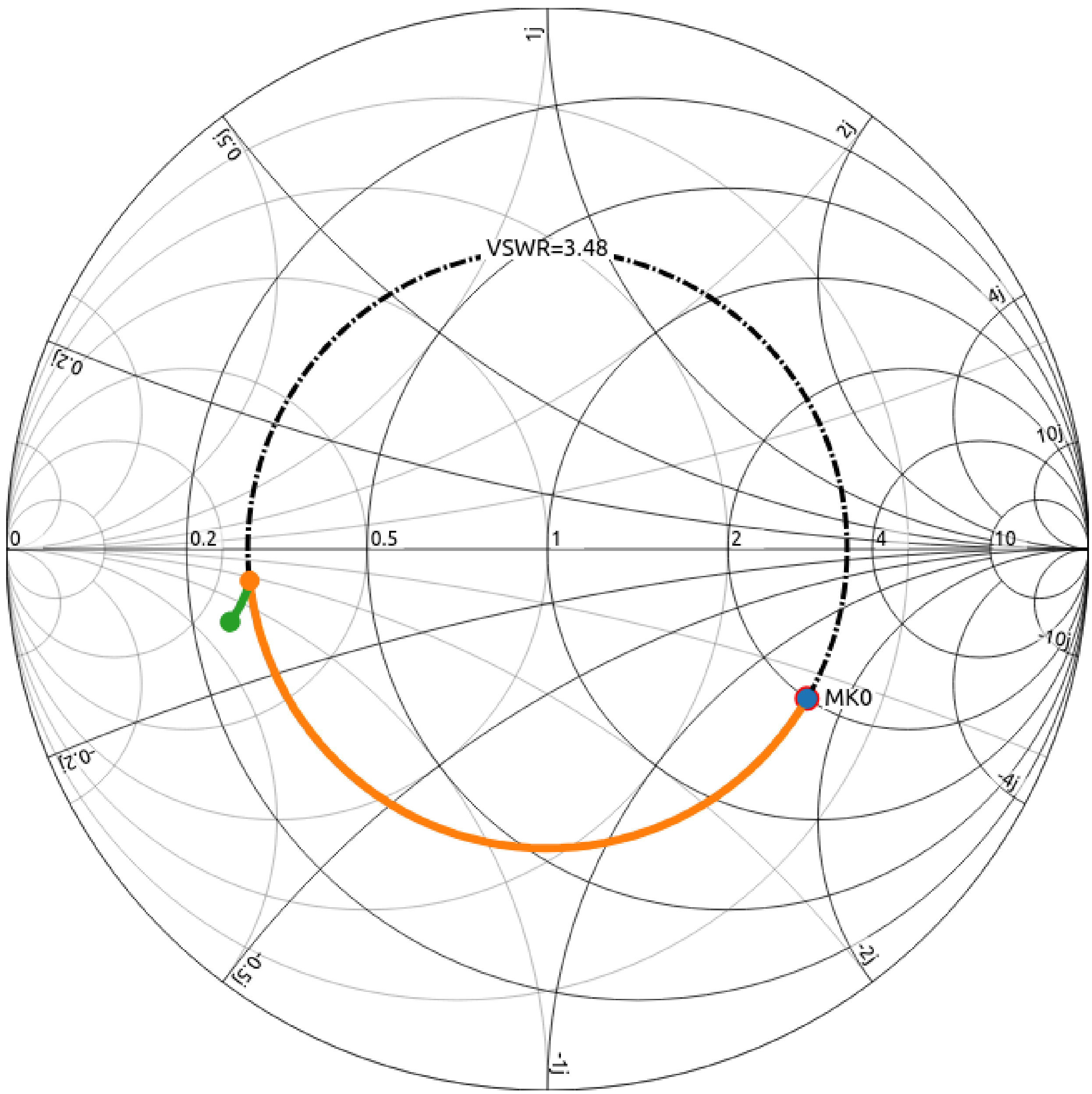
$$z_L = \frac{Z_L}{50} = 2 - j1.59$$

2. Normalized capacitance for a 2.7 pF capacitor is: Z' = Z* Z_0= 2.7pF * 50 = 135 pF

Therefore, we draw the circuit equivalent:



The results are:



Values for the final point on the Smith Diagram:

z_A= 0.25-0.11j

Denormalization:
$Z_A = z_A \cdot 50 = 12.5 - j5.5\Omega$

Frequency = 1 GHz
Impedance = 0.25 - 0.11j (0.27 ∠ -22.82°)
Admittance = 3.37 + 1.42j
Refl-Coeff = -0.586 - 0.134j (0.601 ∠ -167.1°)
VSWR = 4.02
Q-Factor = 0.42

(b) Determine, using the Smith chart, the minimum possible line lengths l and l_s (in terms of λ), so that the input of the circuit is matched. The stub termination can be either an open circuit or a short circuit (based on the criterion of minimum l_s). All transmission lines have a characteristic impedance of 50 Ω.

To achieve matching at the circuit input, the total complex input impedance (at a distance l from point A) must be equal to the characteristic impedance of the line, i.e., $Z_{\text{in}}=Z_0$=50 Ω. Therefore, $z_{\text{in}}=\frac{Z_{\text{in}}}{Z_0}$=1. In terms of normalized admittance, this means that: $y_{\text{in}}=\frac{1}{z_{\text{in}}}$=1+j0.

Since the stub is connected in parallel, it is easier to work with normalized admittances y. We have from question (a) that: y_A≈0.55+j1.35

1. Determination of length l

The section of line with length l transforms the admittance y_A as we move toward the source (clockwise). The goal is to reach a point y_l on the constant conductance circle g=1, so that the stub can then cancel out the imaginary part.

1. We locate y_A≈3.3+j1.4 on the Smith chart.
2. We move along the constant VSWR circle (clockwise - "Toward Generator") until we intersect the g=1 circle.
3. The first intersection (for minimum l) occurs at point:
   y_l=1-j1.4
4. Calculation of l:
   - The position of yA on the "Wavelengths Toward Generator" (WTG) scale is approximately 0.23λ.
   - The position of yl on the WTG scale is approximately 0.33λ.
   - l=0.33λ−0.23λ=0.1λ

2. Determination of length ls (Stub)

For perfect matching (yin=1+j0), the stub must provide a purely imaginary admittance y_s that cancels the -j1.4 of y_l:

y_s=j1.4

Now we choose the termination (Short or Open) for minimum ls:

- Short Circuit: We start from point y=∞ (right edge of the horizontal axis). We move clockwise until we find the value j1.4

- Position y=∞ on WTG: 0.250λ.
  - Position y=j1.4 on WTG: ≈0.15λ.
  - ls,SCls=(0.5−0.250)+0.15=0.4λ
- Open Circuit: We start from point y=0 (left edge of the horizontal axis).
  - Position y=0 on WTG: 0.000λ.
  - Position y=j1.4 on WTG: 0.15λ.
  - ls,OC=0.15λ−0.000λ=0.15λ

Result: For the minimum possible length, we choose the open-circuited stub with ls≈0.15λ.

(b) Solution/Verification with tool: https://onlinesmithchart.com

Links:

https://onlinesmithchart.com/?zo=1&frequencyUnit=GHz&frequency=1&zMarkers=2_-1.59&circuit=blackBox_2_-1.59_0__transmissionLine_0.2_%CE%BB_0_1_1__shortedCap_135_pF_0_0_0__transmissionLine_0.0915_%CE%BB_0_1_1__shortedStub_0.4066_%CE%BB_0_1_1&vswrCircles=3.48

https://onlinesmithchart.com/?zo=1&frequencyUnit=GHz&frequency=1&zMarkers=2_-1.59&circuit=blackBox_2_-1.59_0__transmissionLine_0.2_%CE%BB_0_1_1__shortedCap_135_pF_0_0_0__transmissionLine_0.0915_%CE%BB_0_1_1__stub_0.157_%CE%BB_0_1_1&vswrCircles=3.48

1. Determination of l

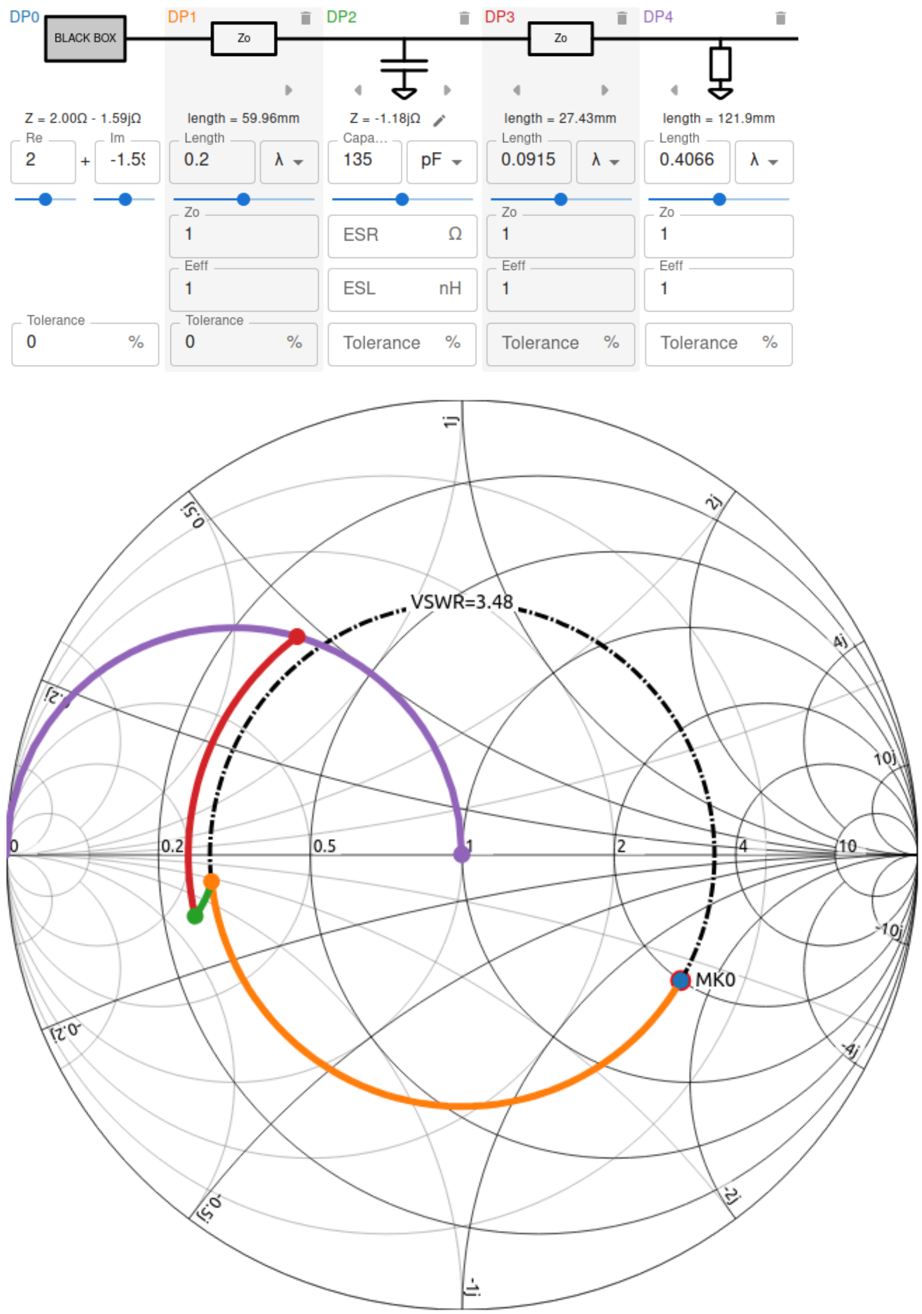
We move along the constant VSWR circle (clockwise - "Toward Generator") until we intersect the g=1 circle (essentially we add a transmission line and test values for λ until yl is obtained with Re(Admittance)=1)

We obtain l = 0.0915λ and intersect g=1 at point yl=1-1.51j

1. Determination of ls:

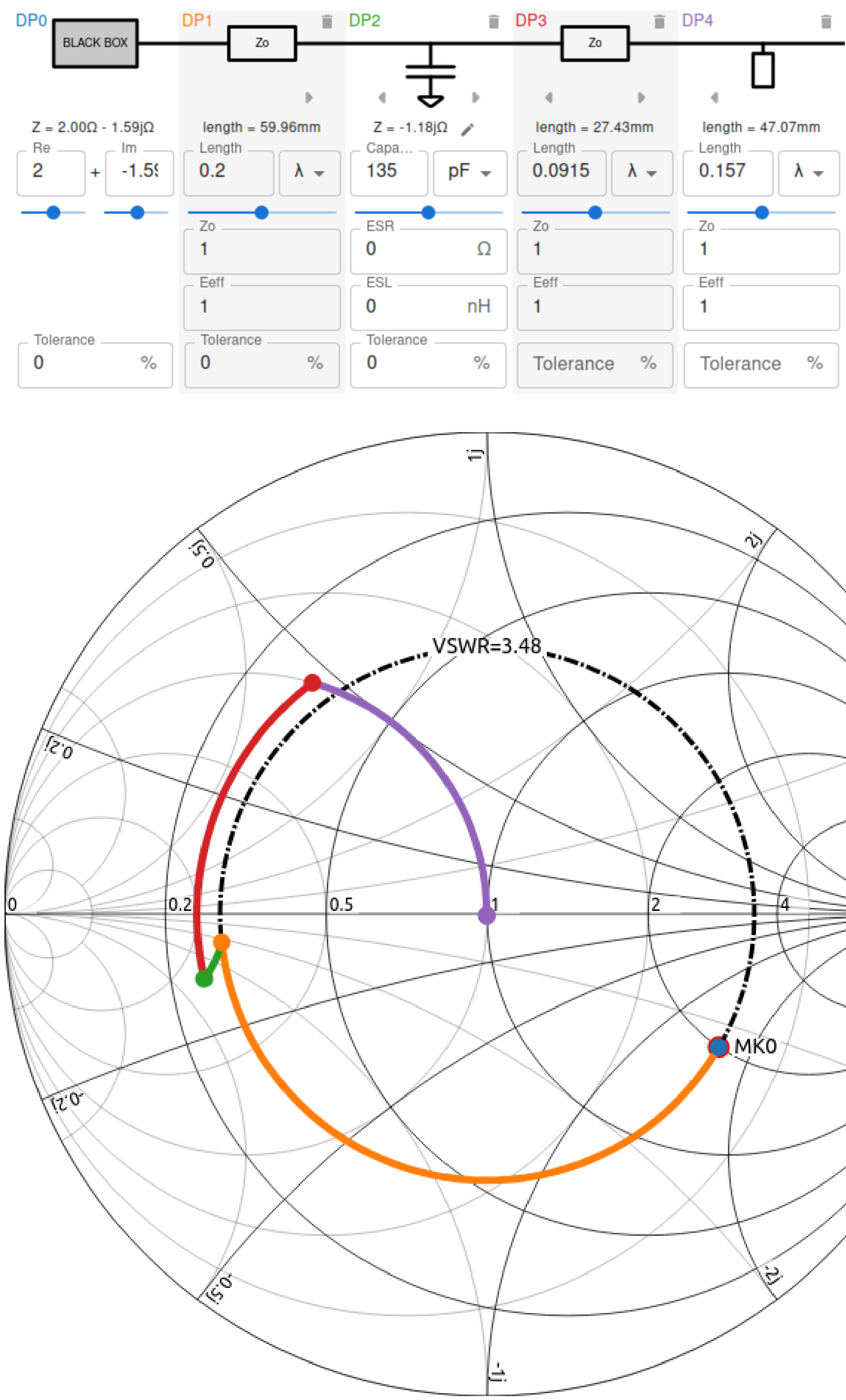
- Short Circuit:
  We add a grounded transmission line in parallel (shorted stub) and increase its length until we reach z=1+0j (i.e., Z=50Ω which is required for the circuit to operate under matching conditions)

Result: ls=~0.4066λ



- Open Circuit:
  We add an ungrounded transmission line in parallel (stub) and increase its length until we reach z=1+0j (i.e., Z=50Ω which is required for the circuit to operate under matching conditions)

Result: ls=~0.157λ



Result: For the minimum possible length, we choose the open-circuited stub with ls≈0.157λ.

(c) Using MATLAB or Python, calculate analytically and make a graph of the magnitude of the reflection coefficient, as a absolute value and in dB, across the frequency band from 0 to 2f0 (at N=201 frequency values). For convenience, first express the propagation constant as a function of frequency (note: the wavelength changes, not the physical length of the lines).

1. Definition of Propagation Constant β(f)

The electrical distance βl changes linearly with frequency. According to the hint:

- β(f)l=0.2 · 2π · f/f0 (for the 0.2λ section)
- β(f)l_match=0.1 · 2π · f/f0 (for the section l=0.1λ)
- β(f)l_s=0.15 · 2π · f/f0 (for the stub)

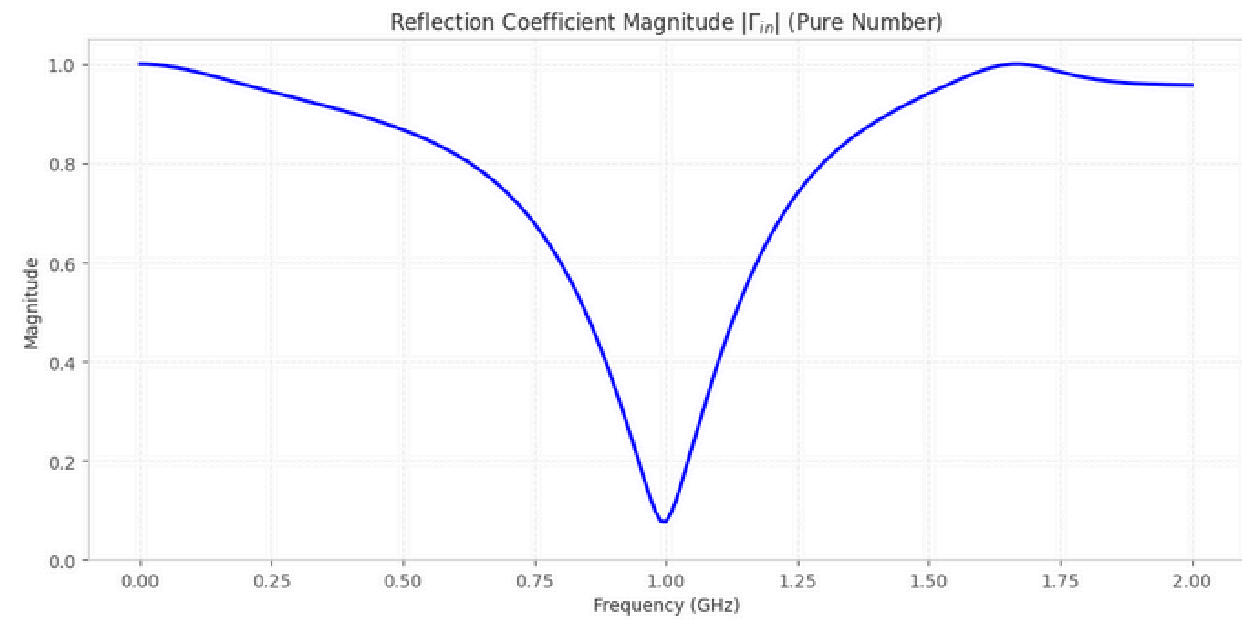
2. Calculation Steps in the Code

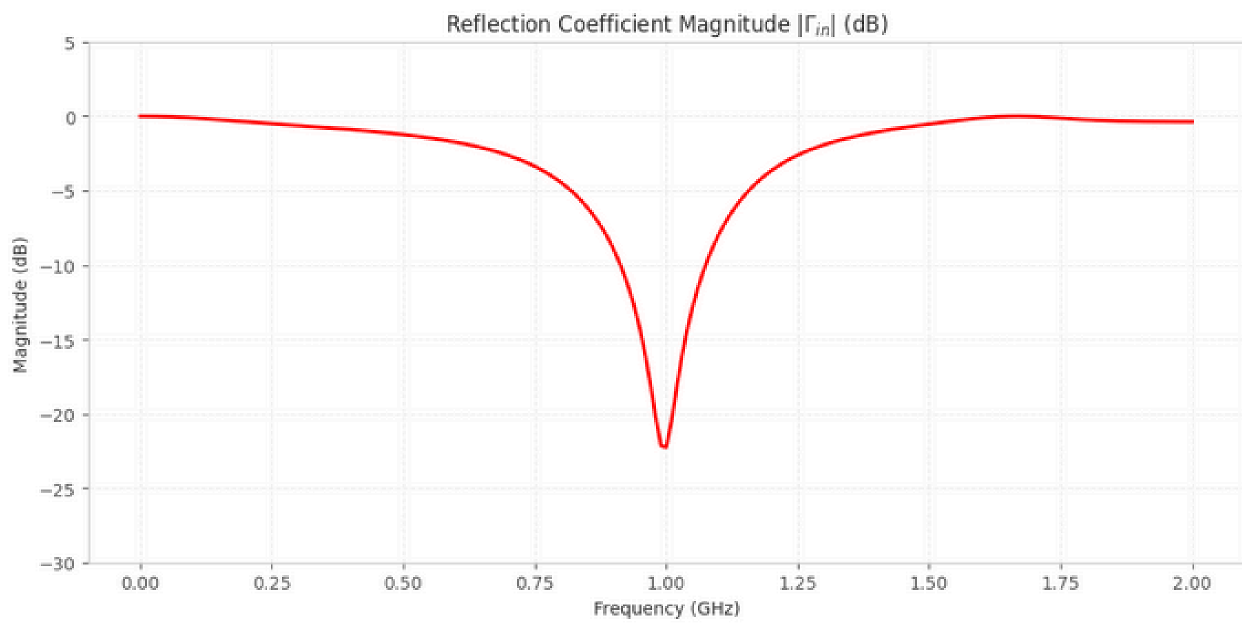
We will use the impedance transformation formula:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$$

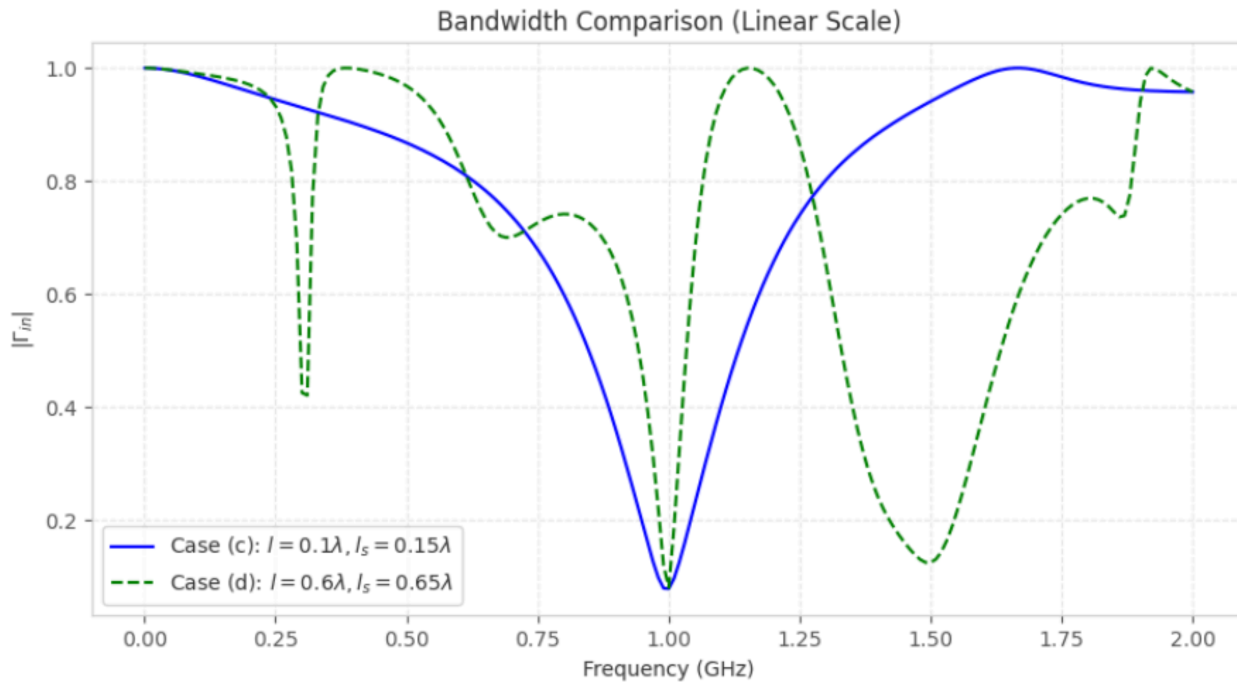
Load: $Z_L(f) = 100 - j\frac{1}{2\pi f(2 \cdot 10^{-12})}$.

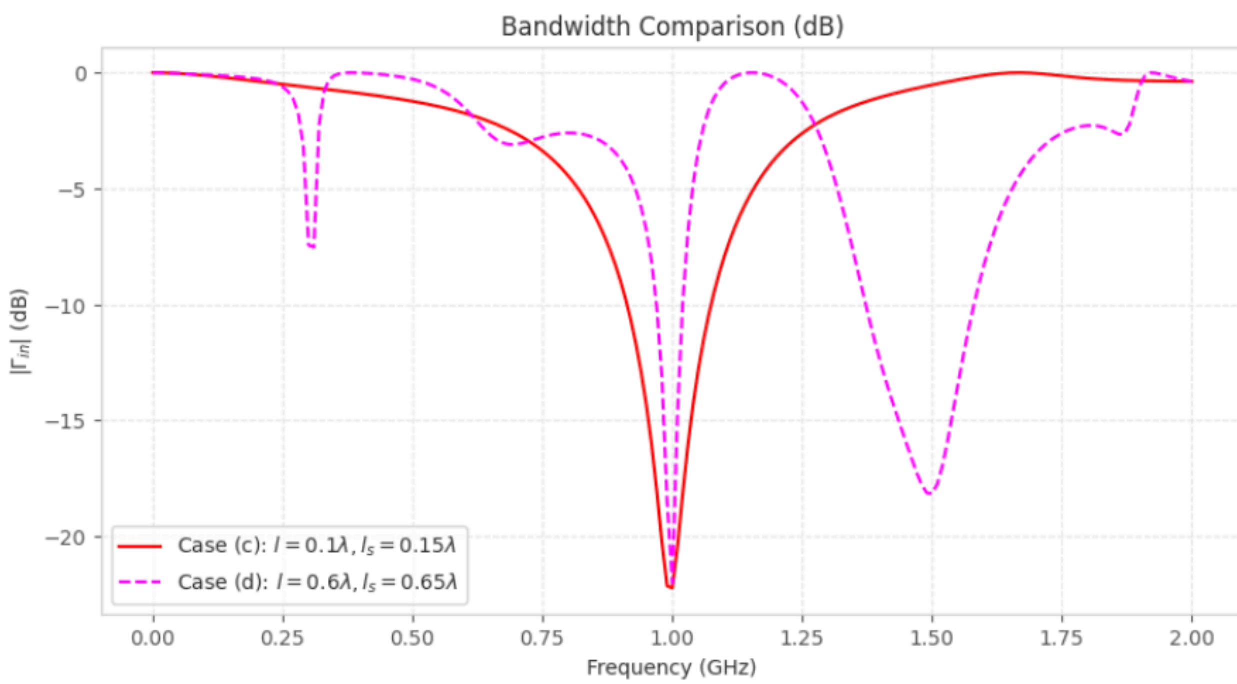
1. Transformation (0.2λ): Find $Z_{line}(f)$ at the point of the first capacitor.
2. Parallel Capacitor (2.7 pF): $Y_A(f) = \frac{1}{Z_{line}(f)} + j2\pi f(2.7 \cdot 10^{-12})$.
3. Transformation (0.1λ): Transform $Z_A = 1/Y_A$ at the stub connection point.
4. Input with Stub: Add the admittance of the OC stub: $Y_{stub}(f) = \frac{j}{Z_0}\tan(\beta(f)l_s)$.
    - Total $Y_{in}(f) = Y_{junc}(f) + Y_{stub}(f)$.
5. Reflection Coefficient: $\Gamma_{in}(f) = \frac{Z_{in}(f) - 50}{Z_{in}(f) + 50}$.





(d) Do the same graph, if we had chosen not the minimum possible but the next possible values for l and ls (and the same stub termination as in c). Compare in terms of bandwidth.





Supplementary Comparison: Periodicity and Selectivity

1. Response Periodicity: It is observed that in case (d) the reflection coefficient exhibits more "dips" (resonances) within the 0–2 GHz range. This is because the longer lines (0.6λ and 0.65λ) reach electrical lengths that are multiples of λ/2 faster as the frequency increases.

2. Useful Bandwidth: Despite the reappearance of matching at other frequencies, the useful bandwidth around the design frequency ($f0=1$ GHz) is significantly smaller. The curve is steeper, making the circuit more "narrowband" and sensitive.
3. Practical Significance: While case (d) "rematches" later, in practice this is often considered a disadvantage (parasitic resonances), as it may allow the passage of unwanted harmonic frequencies.
4. Conclusion: The choice of minimum lengths in question (c) is superior, because it offers the maximum possible stability (tolerance) around the operating point and a cleaner response at the remaining frequencies.

(e) Determine, using the Smith chart, the minimum possible line lengths l and ls (in terms of λ) and the appropriate termination method, for maximum power transfer to the load, if the power supply was from a source with internal complex impedance Zg = 15-j30 Ω.

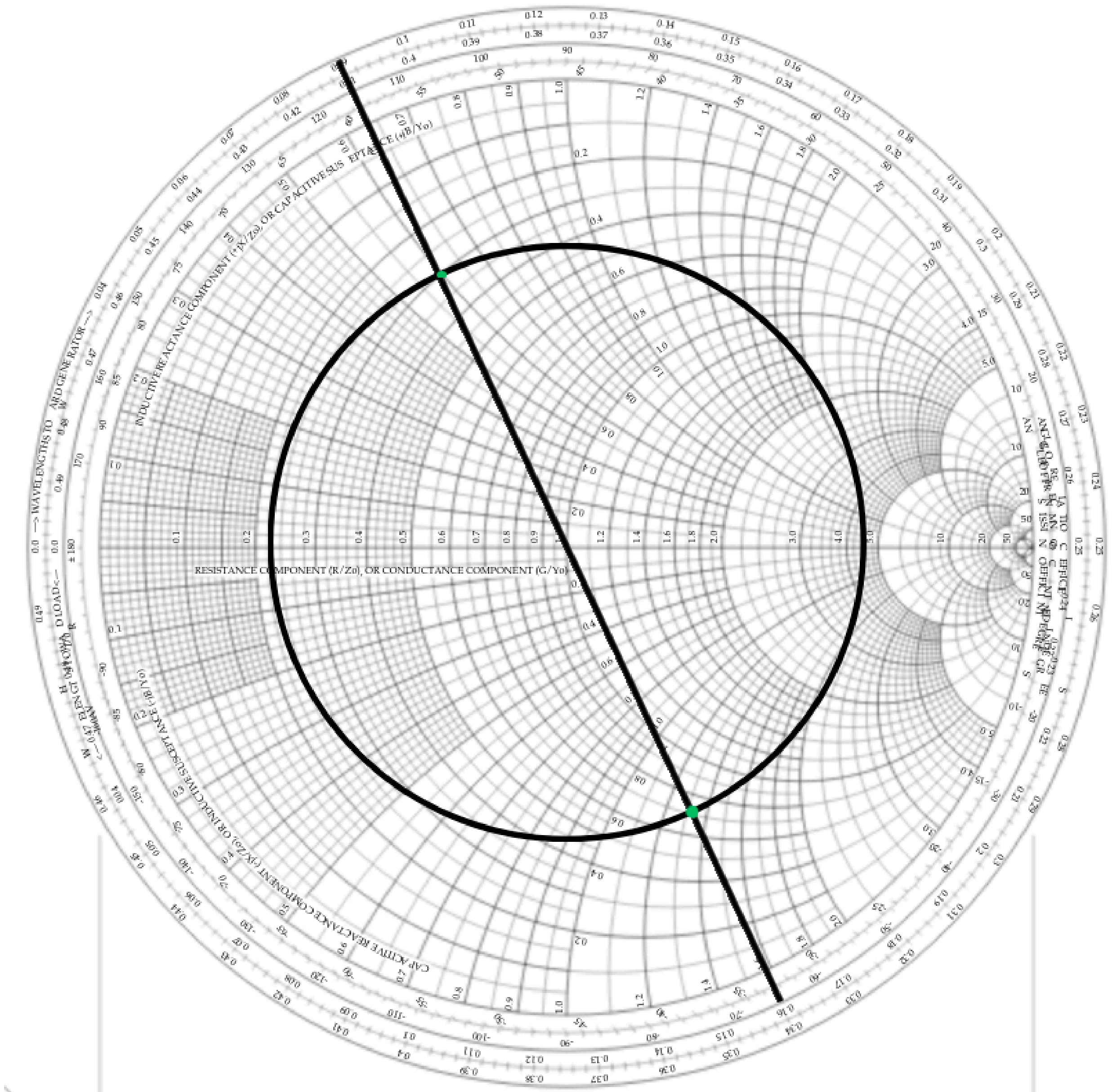
The source has an internal complex impedance $Zg=15-j30\ \Omega$. For maximum power transfer, the input impedance of the circuit Zin (at the point where the source is connected) must be the conjugate of Zg:

$Zin=Zg*=15+j30\ \Omega$

1. Target Normalization

We normalize with respect to $Z_0=50\ \Omega$:

- $zin=5015+j30=0.3+j0.6$
- We find the corresponding admittance yin on the Smith chart (diametrically opposite point): $yin\approx0.7-j1.32$

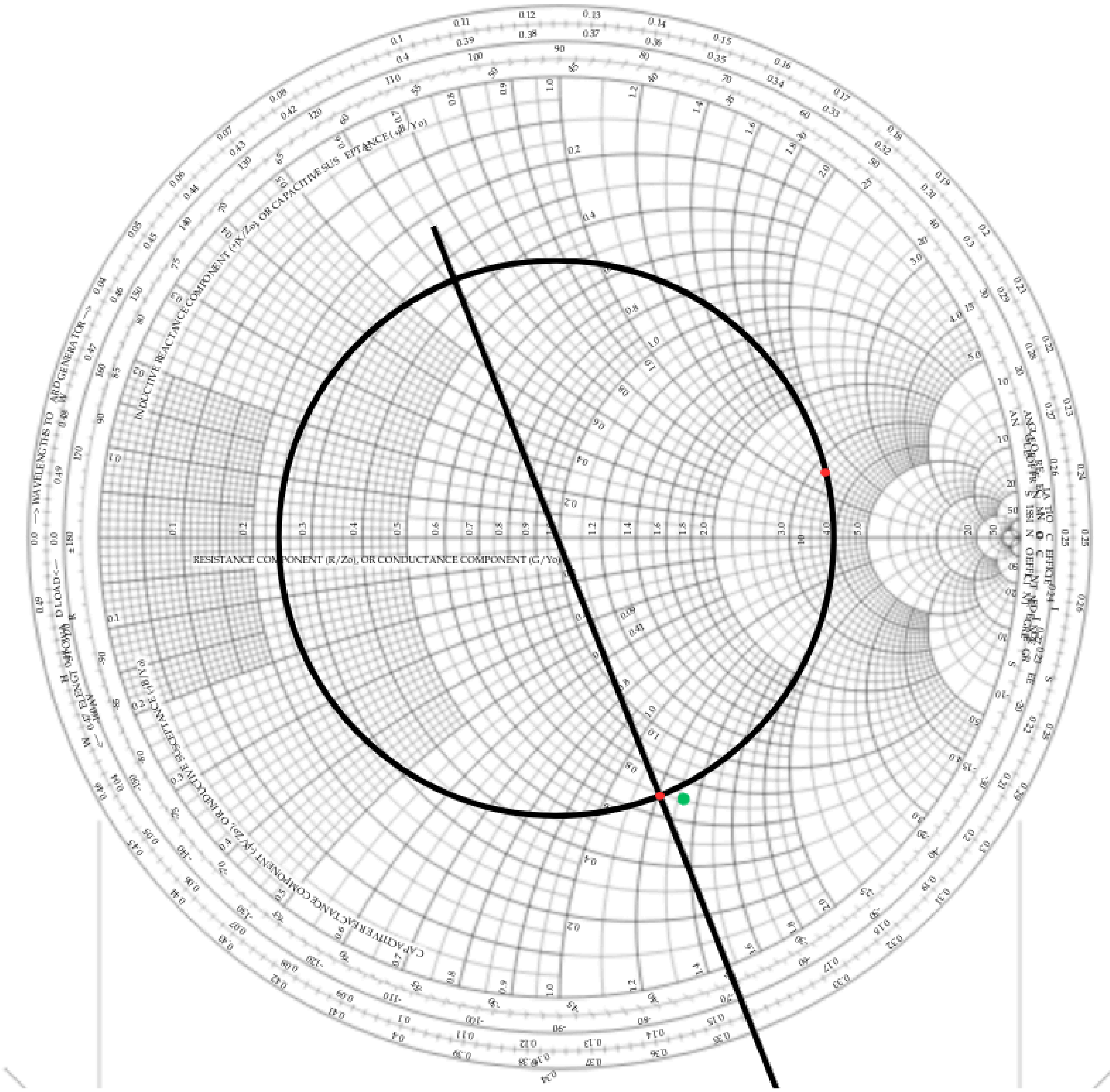


2. Determination of length l

We start from point yA that we calculated in question (a):

- yA≈3.3+j1.4
- Position yA on WTG (Wavelengths Toward Generator): ≈0.23λ.

We rotate clockwise (toward the generator) on the constant VSWR circle until we meet the constant conductance circle g that passes through yin (i.e., the circle g=0.7).



- Intersection point yl=0.7-j1.2
- New WTG position: ≈0.345λ.
- Length l=0.345λ−0.231λ=0.114λ.

3. Determination of length ls (Stub)

Now, the stub must add the necessary susceptance b_s so that from yl=0.7+j1.2 we reach the target yin=0.7−j1.32:

- -j1.2+jb_s=−j1.32⟹b_s=−j0.12.
- Since the required admittance is negative (inductive), we need a short-circuited stub for minimum length.

- Starting point for SC: y=∞ (WTG = 0.250λ).
- We move toward point −j0.12 (WTG ≈0.481λ).
- Length ls=0.481λ−0.250λ=0.231λ.

Summary of Results:

- Termination Method: Short Circuit for minimum ls.
- Length l: Calculated from the WTG position difference between yA and the intersection with the g=0.7 circle.
- Length ls: Calculated to change the susceptance from the line value to the target value of -1.32.

(e) Solution/Verification with tool: https://onlinesmithchart.com

https://onlinesmithchart.com/?zo=1&frequencyUnit=GHz&frequency=1&zMarkers=2_-1.59__0.3_0.6&circuit=blackBox_2_-1.59_0__transmissionLine_0.2_%CE%BB_0_1_1__shortedCap_135_pF_0_0_0__transmissionLine_0.11556_%CE%BB_0_1_1__shortedStub_0.226_%CE%BB_0_1_1

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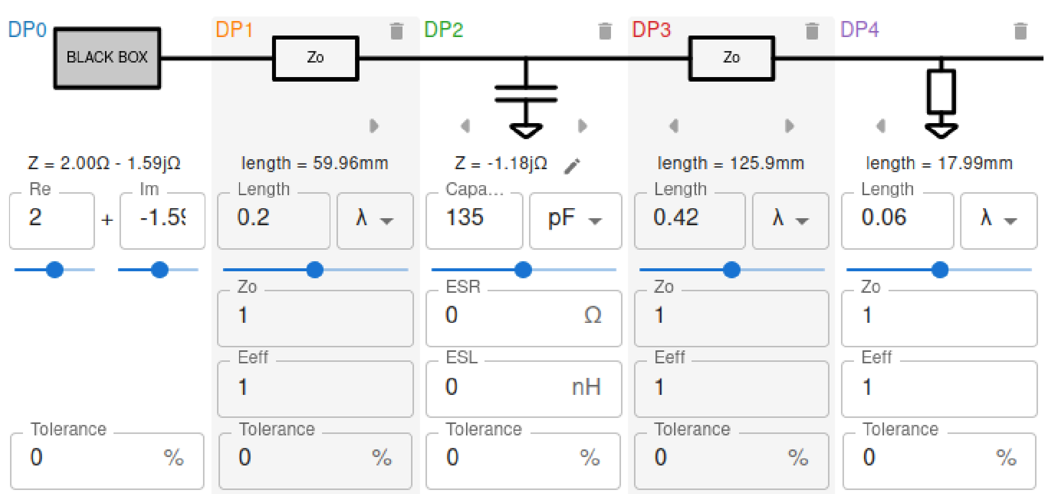
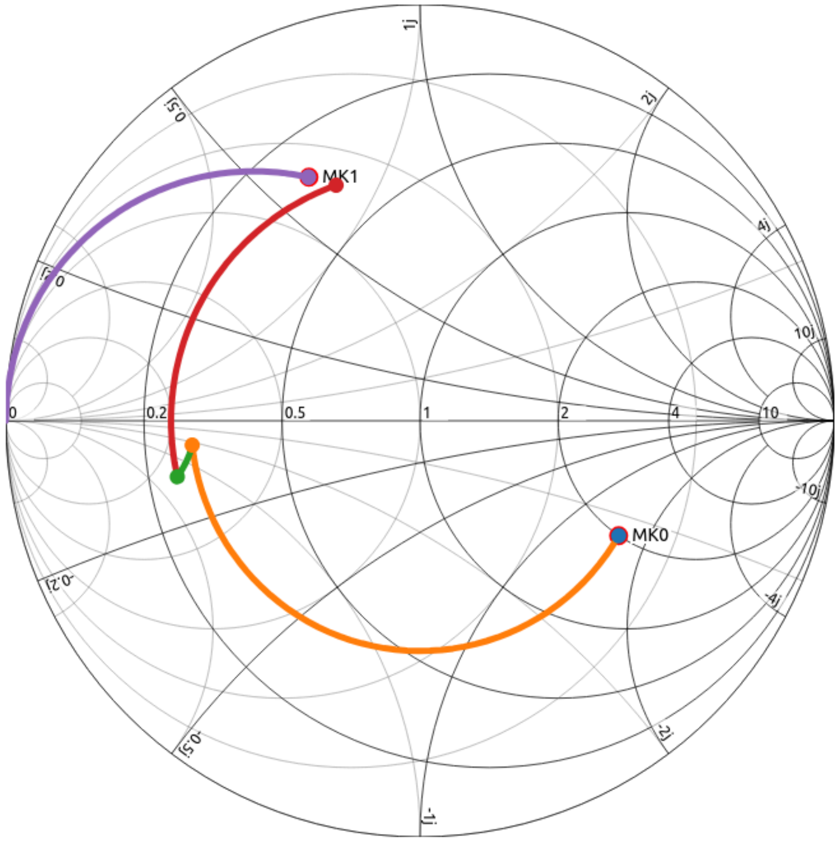
https://onlinesmithchart.com/?zo=1&frequencyUnit=GHz&frequency=1&zMarkers=2_-1.59__0.3_0.6&circuit=blackBox_2_-1.59_0__transmissionLine_0.2_%CE%BB_0_1_1__shortedCap_135_pF_0_0_0__transmissionLine_0.42_%CE%BB_0_1_1__shortedStub_0.06_%CE%BB_0_1_1

From the above, it appears that the solution with the smallest l,ls is: l=0.1155λ, ls = 0.226λ (the first one)





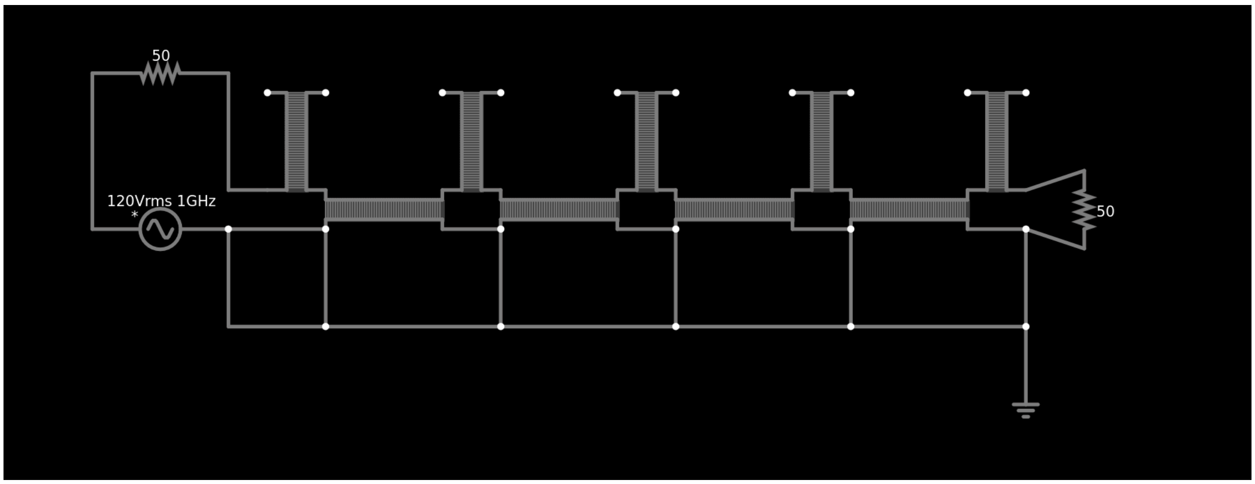
1.2 Transmission line circuits (filters): (a) study, (b) design with optimization

(a) A filter is, as known, a two-port circuit that ensures the passage of the signal in some frequency regions and cut-off in others. Such elements exist practically in every microwave telecommunication system and can be designed using classical techniques from analog filter theory. However, their implementation is not always or only with conventional circuit elements but also with stubs or other types of transmission line sections, as seen in the following example:

The figure shows a practical implementation of the filter in a microstrip circuit (top view) where the metallic elements on the upper side of the dielectric plate are visible. In this example all the transmission line sections, except for the input and output transmission lines, whose length is not specified, have a length of λ/8 at a frequency of 1 GHz (we assume that the microstrip is practically TEM, i.e., the phase velocity is independent of frequency and known). The characteristic impedance of each segment g. m. is noted in the figure and has resulted from a suitable complex mathematical process (see

chapter 11, example 11.7). The circuit is terminated in a matched load (50 Ω). The branched transmission line sections are open-circuited.
Draw the transmission line equivalent circuit of the above circuit and write code to find the reflection coefficient at its input. Specifically, plot the magnitude of the reflection coefficient (in dB) at the input of the circuit in the frequency bands 0-2 GHz. Reflection coefficient values below -50 dB should be set to -50 dB. Conclude what type of filter it is.

Transmission line circuit equivalent:



From falstad (link: https://is.gd/QkOZ6L)
To calculate the line delay, I have:

The electrical length of the line is λ/8 at frequency f=1 GHz.

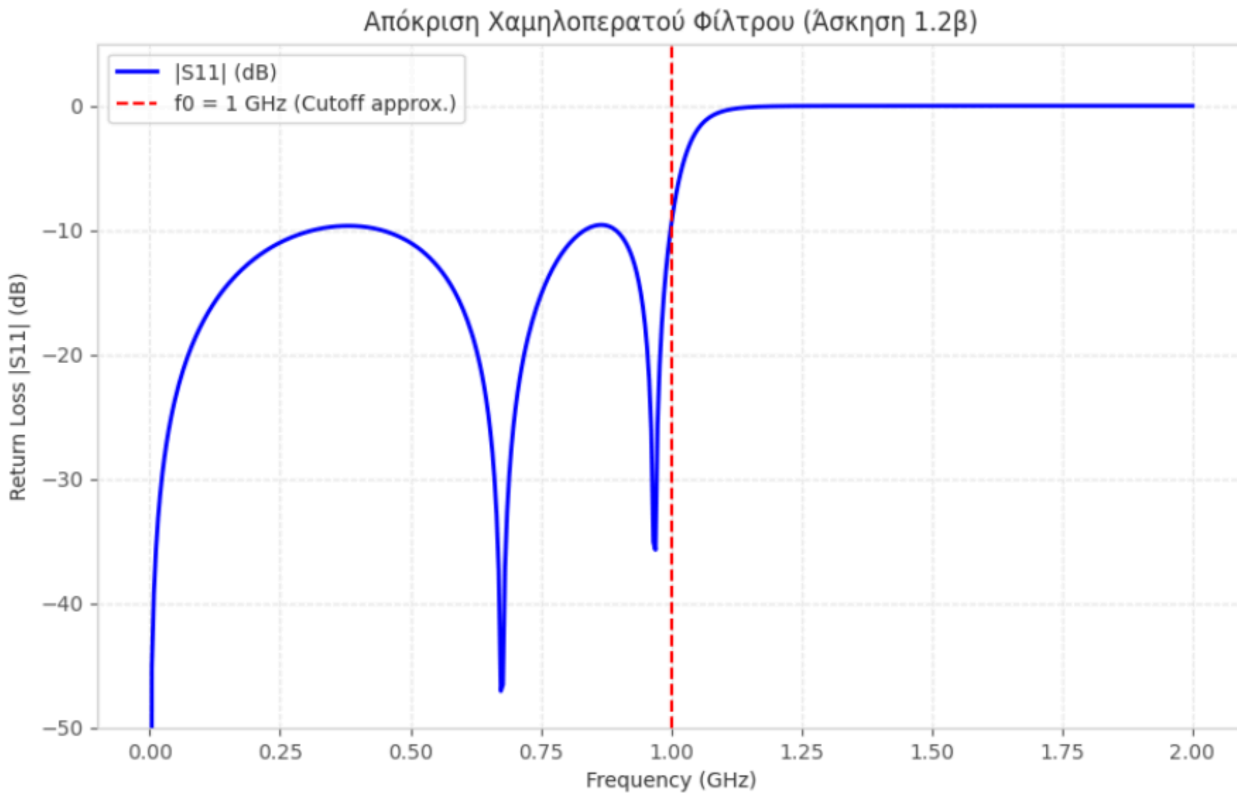
1. The period of the signal is T=1/f=1/1G=1 ns.
2. The length λ corresponds to a delay time of 1 ns.
3. Therefore, λ/8 corresponds to 1/8 ns=0.125 ns=125 ps.

So delay=125p

Each transmission line has an impedance corresponding to the filter, i.e.:

- Horizontal: 81.5 Ω and 80 Ω alternately.
- Vertical: 129.3 Ω (edges), 24 Ω, 19.7 Ω (middle).

Reflection coefficient at the input:



Response Graph Commentary (S11)

The graph depicts the input reflection coefficient (|S11|) as a function of frequency. The main observation points are:

- Passband: In the range from 0 to 1 GHz, the value of |S11| remains consistently below −10 dB. This suggests excellent impedance matching, as less than 10% of the power is reflected back to the source. The "nulls" observed at 0.65 GHz and 0.95 GHz correspond to the optimal matching frequencies for this particular filter.
- Cutoff Frequency: Exactly at the design frequency f0=1 GHz (red dashed line), an abrupt transition of the curve is observed. |S11| rises rapidly toward 0 dB, signaling the end of the passband.
- Stopband: For frequencies above 1.1 GHz, the reflection coefficient approaches 0 dB ($|\Gamma| \approx 1$). This means that almost all the incident power is reflected back to the input and is not transferred to the load. This behavior confirms that the arrangement of 5 parallel stubs and intermediate series lines functions effectively as a Low-Pass Filter.

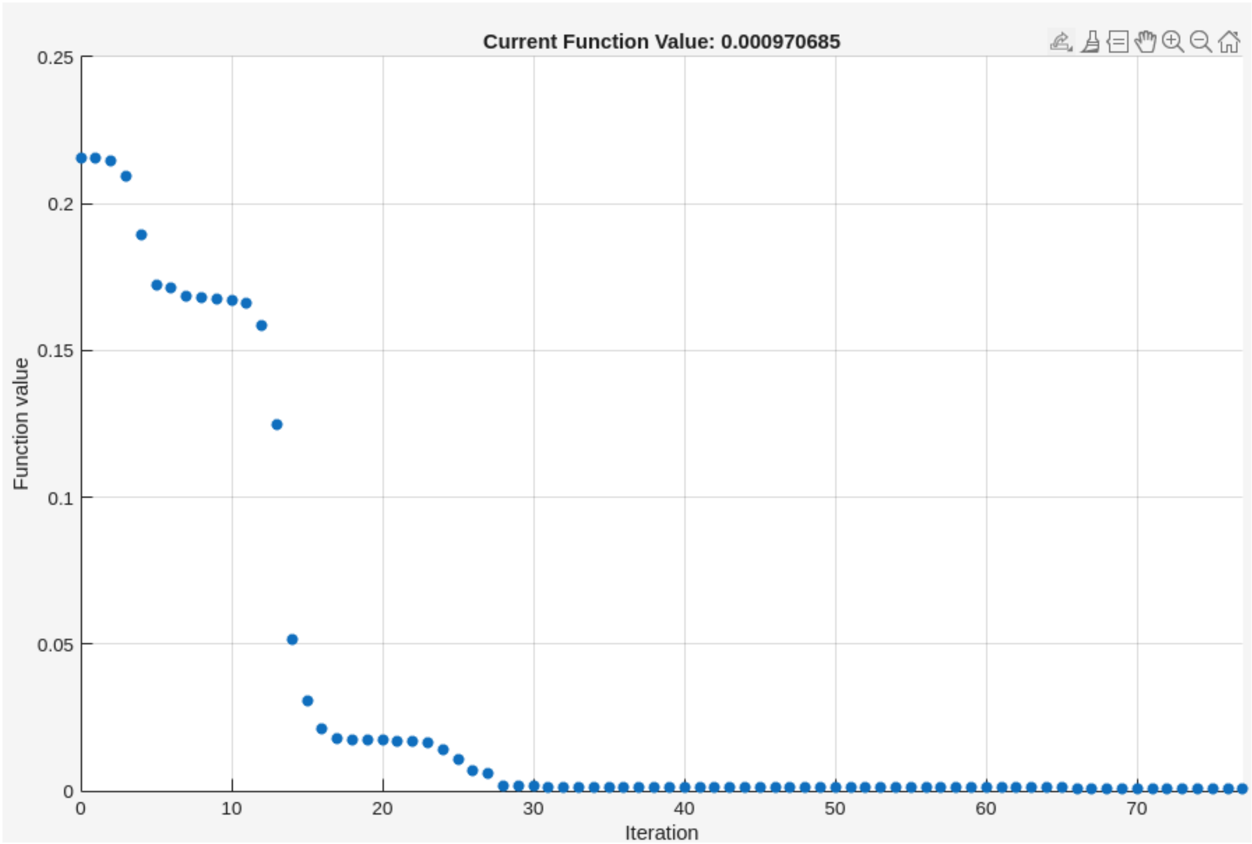
(b) The use of mathematical design techniques is limited to "canonical" problems of simple structure. In most cases this is not feasible and the optimal design requires the application of complex computational optimization techniques. The application of such techniques is a research field of great importance and is considered state-of-the-art in modern design technology in a multitude of technological applications. (....)

- I find the optimal values using fmincon, without the fitness function:

```
>> % Initial estimation (values from part a)
% Format: [Z01, Z02, Z0S1, Z0S2, Z0S3]
x0 = double([81.5, 80.0, 129.3, 24.0, 19.7]);
labels = {'Z01 (Line 1)', 'Z02 (Line 2)', 'Z0s1 (Stub 1)', 'Z0s2 (Stub 2)', 'Z0s3 (Stub 3)'};

% Define Optimization Bounds (Double type to avoid errors)
lb = double([20, 20, 20, 20, 20]);    % Minimum 20 Ohm
ub = double([200, 200, 200, 200, 200]); % Maximum 200 Ohm

% Optimization Options for fmincon
% We use 'sqp' which is very reliable for microwave circuit problems
options = optimoptions('fmincon', ...
    'Display', 'iter-detailed', ...
    'Algorithm', 'sqp', ...
    'PlotFcns', @optimplotfval, ...
    'OptimalityTolerance', 1e-6);

fprintf('\n--- Starting fmincon Optimization (Fitness Function #2) ---\n');

% Calling fmincon with explicit double empty matrices for linear constraints
[solution, fval] = fmincon(@Filter1, x0, ...
    double.empty(0,5), double.empty(0,1), ... % Linear inequality A*x <= b
    double.empty(0,5), double.empty(0,1), ... % Linear equality Aeq*x = beq
    lb, ub, [], options);

%% Results Display
fprintf('\n==================================================\n');
fprintf('         OPTIMIZATION RESULTS SUMMARY\n');
fprintf('==================================================\n');
fprintf('%-15s | %-12s | %-12s\n', 'Parameter', 'Initial (a)', 'Optimized (b)');
fprintf('--------------------------------------------------\n');

for i = 1:length(x0)
    fprintf('%-15s | %-12.2f | %-12.2f Ohm\n', labels{i}, x0(i), solution(i));
end

fprintf('--------------------------------------------------\n');
fprintf('Final Fitness Value: %.6f\n', fval);
fprintf('==================================================\n');
```

```
function S = Filter1(p)
    % p = [Z01, Z02, Z0s1, Z0s2, Z0s3]
    Z01 = p(1); Z02 = p(2); Z0s1 = p(3); Z0s2 = p(4); Z0s3 = p(5);
    f0 = 1e9; Z0_sys = 50;

    f = 0.01e9:0.01e9:2e9;
    N = length(f);
    betaL = (pi/4) .* (f./f0);

    % Υπολογισμός με ABCD Matrices για σταθερότητα
    M_line = @(Z0, th) [cos(th), 1j*Z0*sin(th); 1j*sin(th)/Z0, cos(th)];
    M_stub = @(Z0, th) [1, 0; 1j*tan(th)/Z0, 1];

    S11_abs = zeros(1, N);
    for k = 1:N
        th = betaL(k);
        T = M_stub(Z0s1,th)*M_line(Z01,th)*M_stub(Z0s2,th)*M_line(Z02,th)*...
            M_stub(Z0s3,th)*M_line(Z02,th)*M_stub(Z0s2,th)*M_line(Z01,th)*...
            M_stub(Z0s1,th);
        A = T(1,1); B = T(1,2); C = T(2,1); D = T(2,2);
        S11 = (A + B/Z0_sys - C*Z0_sys - D) / (A + B/Z0_sys + C*Z0_sys + D);
        S11_abs(k) = abs(S11);
    end

    % Χωρισμός ζωνών
    S11_pass = S11_abs(1:N/2); % 0 - 1 GHz

    % FITNESS FUNCTION #1: Μόνο ελαχιστοποίηση στη ζώνη διάβασης
    S = mean(S11_pass);

    S = real(double(S));
end
```

Results:



==

OPTIMIZATION RESULTS SUMMARY

==

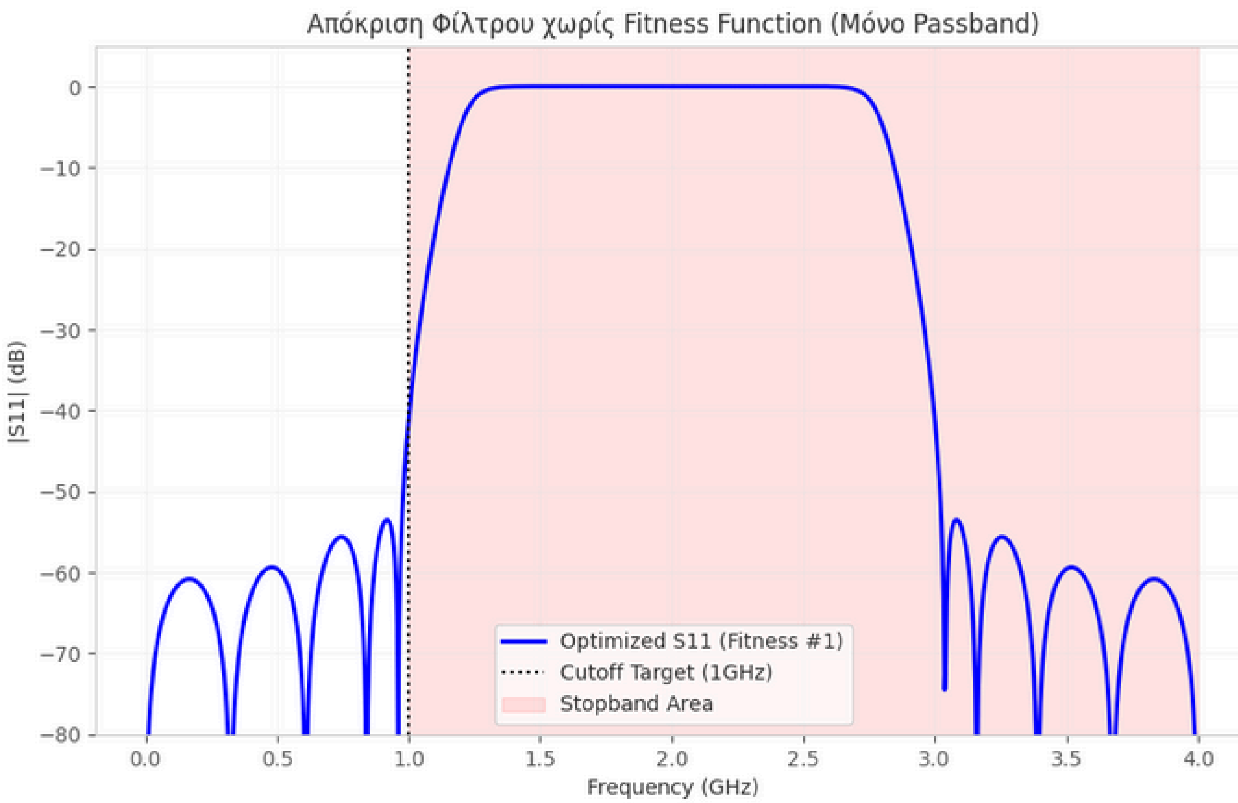
Parameter | Initial (a) | Optimized (b)

Z01 (Line 1) | 81.50 | 76.45 Ohm
Z02 (Line 2) | 80.00 | 93.68 Ohm
Z0s1 (Stub 1) | 129.30 | 129.24 Ohm
Z0s2 (Stub 2) | 24.00 | 43.49 Ohm
Z0s3 (Stub 3) | 19.70 | 36.28 Ohm

Final Fitness Value: 0.000971

==

I am using these parameters to run the python script from question 1.2a again:



Graph Commentary

Despite the excellent performance in the passband, the first optimization is judged technically unacceptable for the following reasons:

1. Stopband Leakage: It is observed that after 2.7 GHz, the reflection coefficient ∣S11∣ shows an abrupt and unpredictable drop, returning to values close to -60 dB. This creates a "passband window" (spurious passband), where the filter allows full power passage instead of reflecting it.
2. Inability to Suppress the 3rd Harmonic: In RF engineering, it is critical that a filter cutting at 1 GHz maintains high rejection at least up to 3 GHz. The 3rd harmonic ($3 \times fc$) is the most common source of interference from non-linear elements (such as amplifiers). The specific graph proves that the filter fails to "kill" these interferences, making the system vulnerable to noise.
3. Richards Periodicity Effects: This collapse is a result of the periodic nature of the $L=\lambda/8$ transmission lines. As the electrical angle θ approaches 180◦ (at 4 GHz), the response tends to repeat the DC behavior. Without the Fitness Function #2 constraint, the algorithm "sacrificed" the rejection in the 3rd harmonic region to achieve a redundant (and practically unnecessary) perfection in the passband.

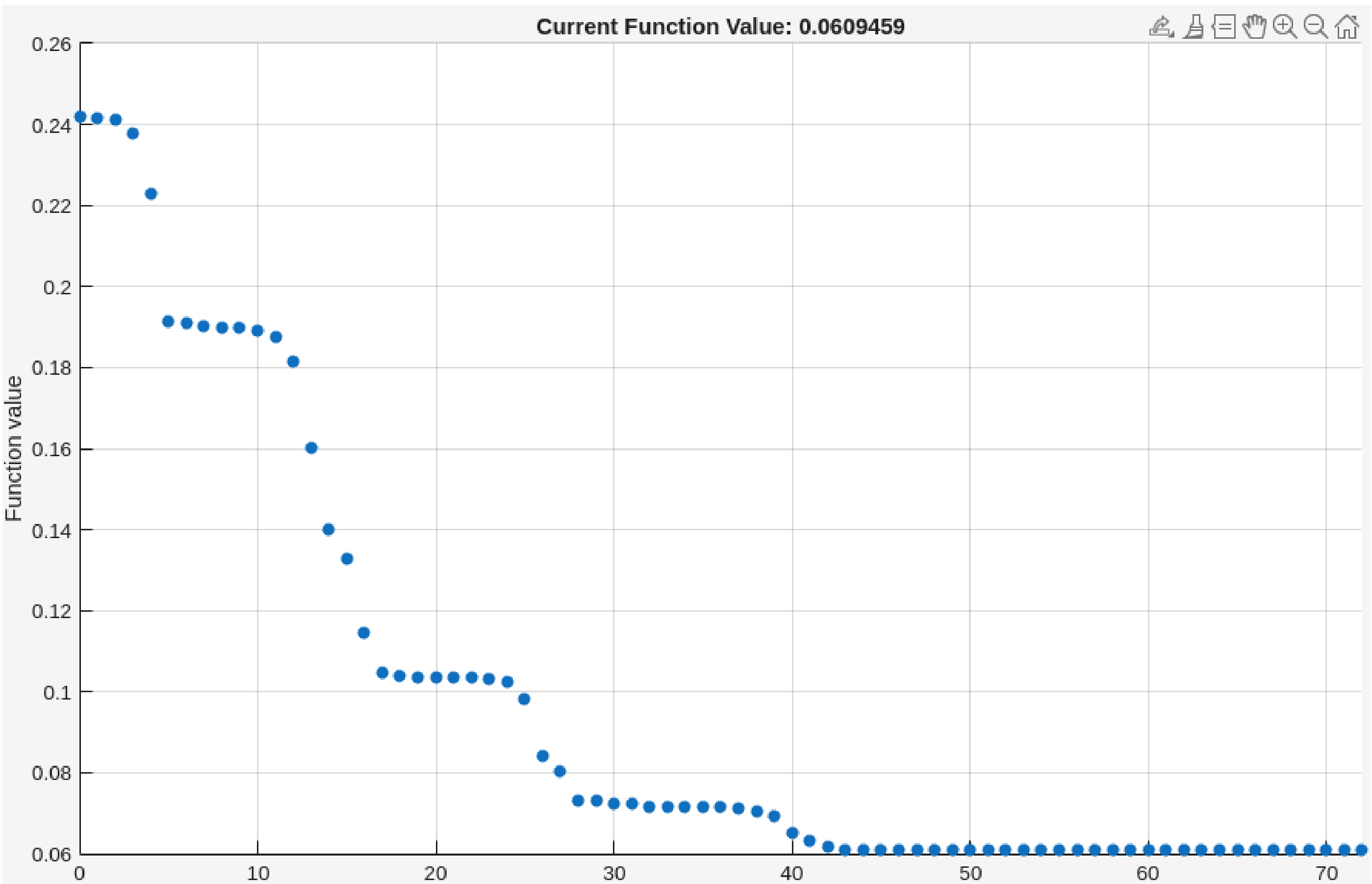
Conclusion

The first design is the result of "over-optimization" on a single goal. For the filter to be functional in real conditions, the objective function must include a penalty term that forces ∣S11∣ to remain close to unity (0 dB) at least up to 3 GHz.

Solution: implementing fitness function in the Filter1 file:

```
% Fitness Function #2:
% Minimize mean reflection in passband + Distance from 1.0 in stopband
S = mean(S11_pass) + (1 - mean(S11_stop));

% Ensure S is a real double scalar
S = real(double(S));
```

Results:



==

OPTIMIZATION RESULTS SUMMARY

==

Parameter | Initial (a) | Optimized (b)

Z01 (Line 1) | 81.50 | 83.72 Ohm

Z02 (Line 2) | 80.00 | 112.36 Ohm

Z0s1 (Stub 1) | 129.30 | 128.88 Ohm

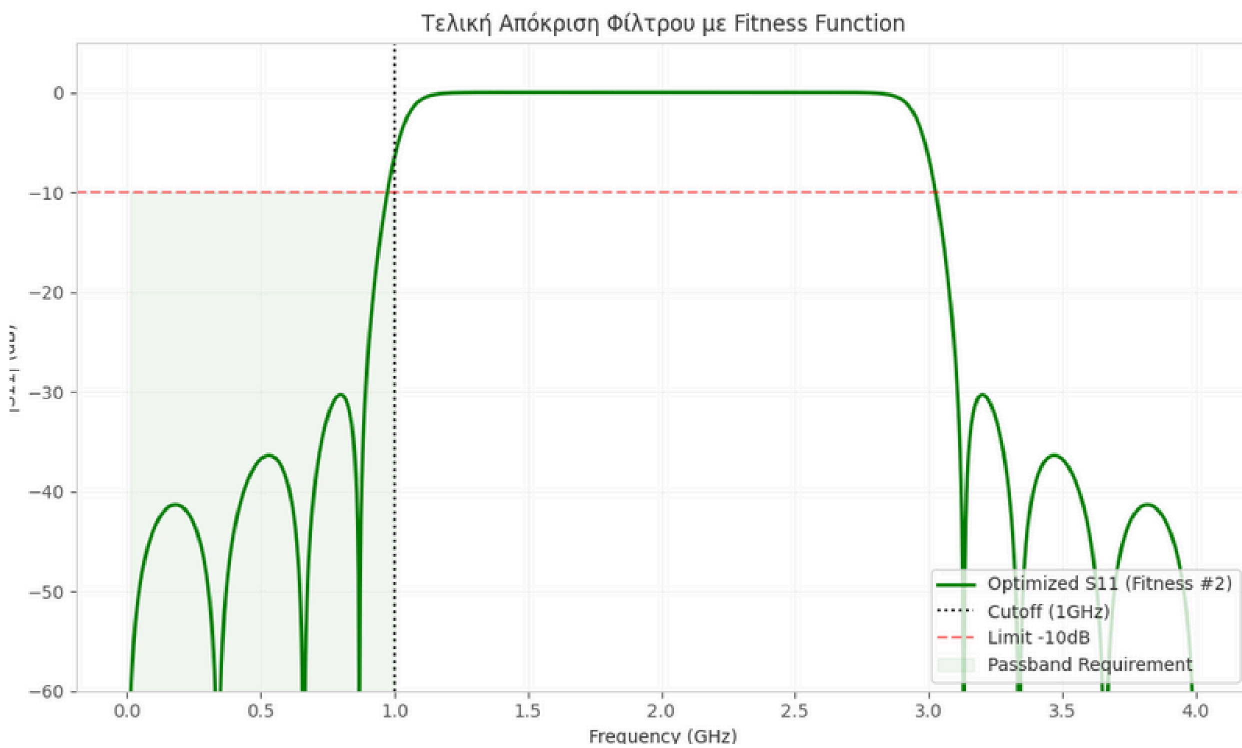
Z0s2 (Stub 2) | 24.00 | 33.59 Ohm

Z0s3 (Stub 3) | 19.70 | 27.49 Ohm

Final Fitness Value: 0.060946

==

I am using these parameters to run the python script from question 1.2a again:



Graph Commentary with Fitness Function

The application of the complex cost function (Fitness Function) resulted in a significantly more balanced and reliable response, which fully satisfies the technical specifications of the project:

- Compliance and Performance (Passband Performance): As indicated by the green shaded region, the reflection coefficient ∣S11∣ is maintained consistently below the limit of -10 dB across the entire passband (0−1 GHz). The values achieved (below -30 dB) ensure that more than 99.9% of the incident power is transferred to the load, minimizing return losses.

- Stopband Stability: Unlike the initial approach, the curve now remains "tangent" to the 0 dB level from 1.2 GHz up to 3 GHz. The elimination of "passband windows" (spurious passbands) in the 3rd harmonic region is crucial, as it ensures that the filter effectively rejects interference and harmonic noise that could disrupt system operation.
- Roll-off: Extremely abrupt transition exactly at the cutoff frequency (1 GHz), approaching the ideal "brick-wall" response. This high selectivity allows for effective separation of adjacent frequency channels, reducing adjacent channel interference.

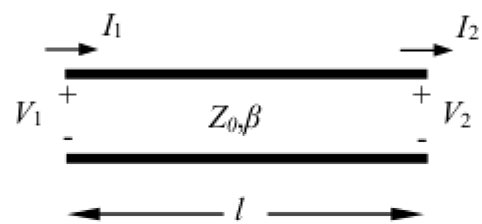
Concluding Comparison

The optimization with the Fitness Function proves superior, as it "forced" the algorithm to sacrifice the redundant (and practically unnecessary) matching of the order of -60 dB from the first trial, in order to guarantee the robustness of the stopband. The result is a realistic, high-quality telecommunication filter.

1.3 Analysis of transmission line circuits with non-“tree structure”

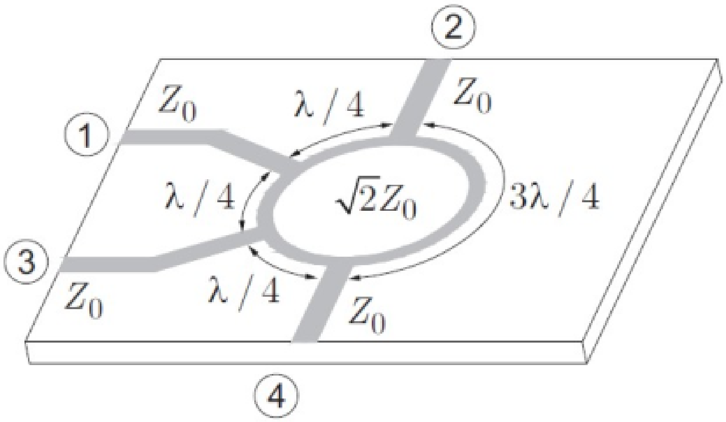
A very large number of microwave circuits have a "tree structure," in the sense that the microwave generator feeds a transmission line, which branches into others, which in turn can branch etc. without other connections between branches (the circuits of exercises 1.1 and 1.2 belong to this category). The analysis of these circuits can be done, as is known, from the loads to the input, finding successively the input impedances of individual lines with known loads and applying simple Kirchhoff laws (e.g. connecting resistors in series or in parallel) at the connection points. This method does not apply to circuits with non-tree structure. In order to analyze such a circuit, we use the relationship connecting voltages and currents at the input and output of a transmission line.

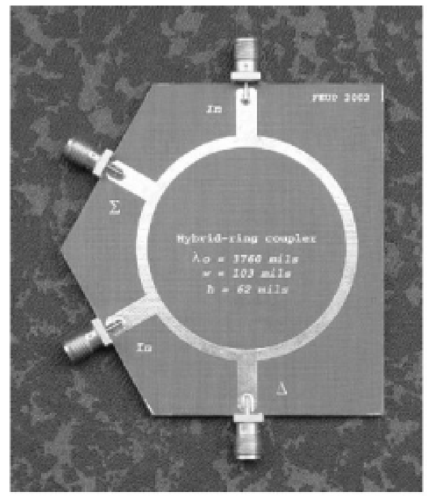
$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} \cos\beta l & jZ_0 \sin\beta l \\ \dfrac{j}{Z_0}\sin\beta l & \cos\beta l \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}.$$



(a) Using the above relationship, write the equations for calculating the input resistance and the reflection coefficient (as a function of frequency) for the following four-port microstrip circuit, as seen in side view and in physical implementation.

Specifically: the input of the circuit is (1), while its outputs are (2), (3) and (4), which feed matched loads. Consider port (1) exactly at the point where it branches into the two λ/4 lines. At this point consider feeding with a microwave voltage source of 1 Volt (without internal resistance). The transmission lines are microstrips (with return conductor obviously the ground). The supply and connection lines to the outputs are Z0 = 50 Ω and the outputs feed matched loads. The internal transmission lines (in the ring) have lengths shown in the figure, at a frequency of 5 GHz and characteristic impedance 2Z 0 (their circular shape plays no role). Consider ports (2), (3) and (4) exactly at the branching points (as well as (1)) and as variables the voltage and current at the input and output of each transmission line segment. Apply also the additional relationships resulting from Kirchhoff laws at the connection points. Thus formulate a system of equations in the form Ax=b (A square matrix, x the vector of unknowns and b the known right hand side).





The hybrid ring circuit consists of four ports (1, 2, 3, 4) connected via transmission line segments with characteristic impedance $Z_c = \sqrt{2}Z_0$. Due to its closed-loop structure, the solution requires the formulation of a system of linear equations of the form $\mathbf{Ax} = \mathbf{b}$.

A. Definition of Unknowns Vector

The vector of unknowns $\mathbf{x}$ includes the voltages and currents at the connection nodes:
$\mathbf{x} = [V_1, I_1, V_2, I_2, V_3, I_3, V_4, I_4]^T$

B. ABCD Equations for Line Segments

For each line segment between ports i and j, we use the transfer matrix. For sections of length $\lambda/4$ ($\beta l = \pi/2$), it holds:

$$\begin{bmatrix} V_i \\ I_i \end{bmatrix} = \begin{bmatrix} 0 & j\sqrt{2}Z_0 \\ \frac{j}{\sqrt{2}Z_0} & 0 \end{bmatrix} \begin{bmatrix} V_j \\ I_j \end{bmatrix}$$

From the analysis of the four branches of the ring, the equations result:

1. Section 1-2 ($\lambda/4$): $V_1 - j\sqrt{2}Z_0 I_2 = 0$

2. Section 1-2 ($\lambda/4$): $I_1 - \frac{j}{\sqrt{2}Z_0}V_2 = 0$
3. Section 2-3 ($\lambda/4$): $V_2 - j\sqrt{2}Z_0 I_3 = 0$
4. Section 2-3 ($\lambda/4$): $I_2 - \frac{j}{\sqrt{2}Z_0}V_3 = 0$
5. Section 3-4 ($\lambda/4$): $V_3 - j\sqrt{2}Z_0 I_4 = 0$
6. Section 3-4 ($\lambda/4$): $I_3 - \frac{j}{\sqrt{2}Z_0}V_4 = 0$
7. Section 4-1 ($3\lambda/4$): $V_4 + j\sqrt{2}Z_0 I_1 = 0$
8. Section 4-1 ($3\lambda/4$): $I_4 + \frac{j}{\sqrt{2}Z_0}V_1 = 0$

C. Boundary Conditions (Source and Loads)

Port 1 (Input): $V_1 = 1$
Ports 2, 3, 4 (Outputs):
$V_2 - Z_0 I_2 = 0$
$V_3 - Z_0 I_3 = 0$
$V_4 - Z_0 I_4 = 0$

D. Formation of Matrix $\mathbf{A}$ and Vector $\mathbf{b}$

By gathering the above equations in the form $\mathbf{Ax} = \mathbf{b}$, the matrix $\mathbf{A}$ (8×8) is formed as:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & -j\sqrt{2}Z_0 & 0 & 0 & 0 & 0 \\ 0 & 1 & \frac{-j}{\sqrt{2}Z_0} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -j\sqrt{2}Z_0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{-j}{\sqrt{2}Z_0} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & -j\sqrt{2}Z_0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & j\sqrt{2}Z_0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -Z_0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The vector of constant terms is:
$\mathbf{b} = [0, 0, 0, 0, 0, 0, 1, 0]^T$

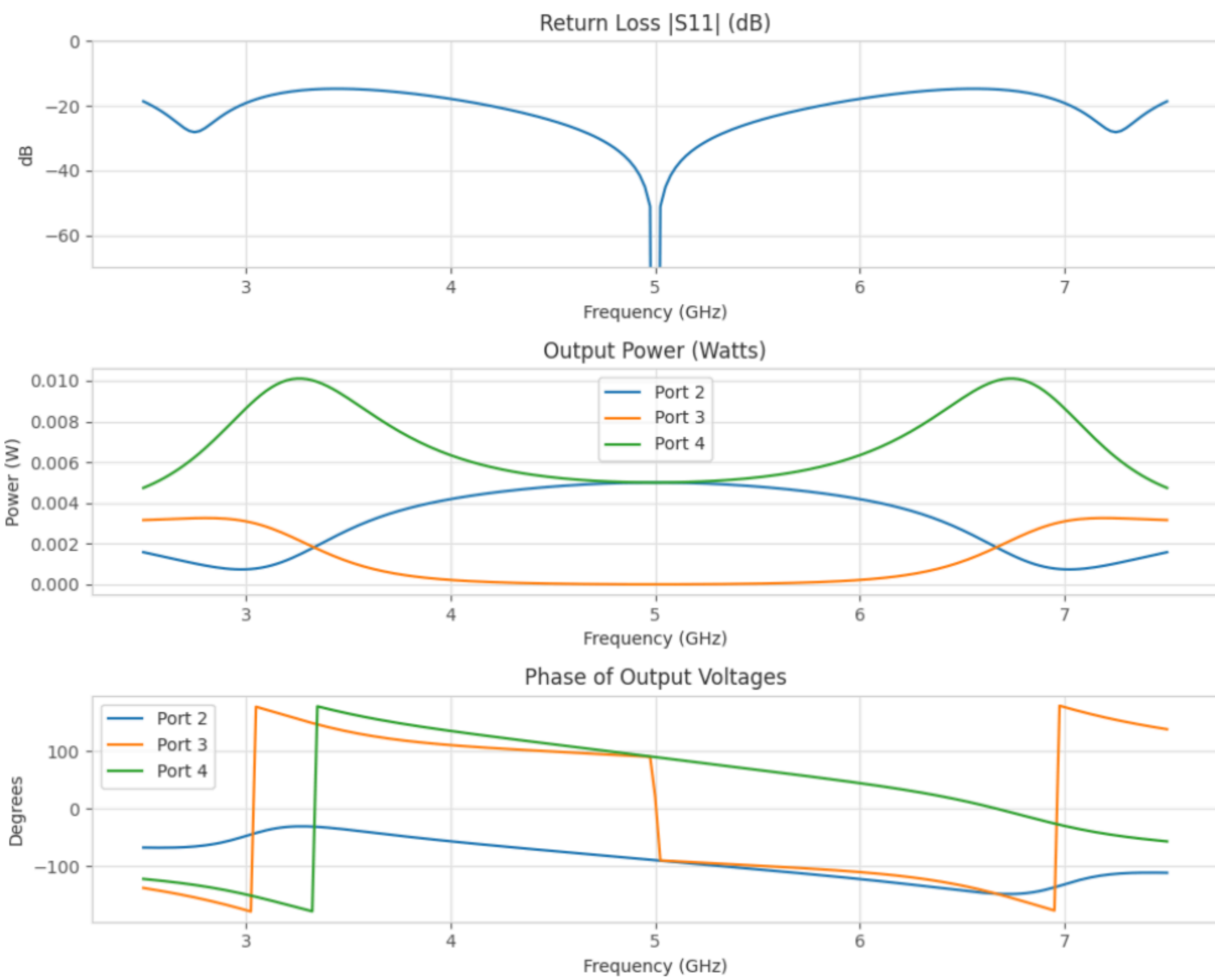
E. Calculation of Input Parameters

After solving the system in MATLAB ($\mathbf{x} = \mathbf{A} \setminus \mathbf{b}$), we calculate:

Input Impedance: $Z_{in} = \dfrac{V_1}{I_1}$

Reflection Coefficient: $\Gamma = \dfrac{Z_{in} - Z_0}{Z_{in} + Z_0}$

b) Solve the system of equations with Matlab for 201 supply frequencies, from 2.5 to 7.5 GHz (with the command x=A\b). Plot the magnitude of the reflection coefficient (in dB) in this band. For each frequency, calculate the power at the input and the power at the three outputs and plot them as a function of frequency in a common diagram. Likewise, plot the magnitude and phase of the voltages at the three outputs. From all the above data determine what is the operation of the circuit and give the range of frequencies of "good operation".



Result Commentary and Circuit Operation

From the analysis of the above diagrams, we find that the circuit operates as a 180 degree Hybrid Ring (Rat-Race Coupler).

The behavior of the circuit is analyzed as follows:

1. Behavior at Design Frequency ($f0=5$ GHz):

- Excellent Matching (Return Loss): In the first diagram we observe a deep "dip" in the reflection coefficient $|S11|$ at 5 GHz (below -40 dB). This suggests that at the design frequency the circuit is perfectly matched and practically all power enters the ring.
- Equal Power Split: In the second diagram (Output Power), we see that at 5 GHz the power curves for Port 2 (blue) and Port 4 (green) intersect at exactly the same point (0.005 W). Given that the source (1V in 50Ω) provides a total available power of 0.01 W, the circuit functions as an ideal 3dB splitter, dividing the power exactly in half.
- Perfect Isolation: At the same frequency, the power at Port 3 (orange line) becomes zero. The signal reaches port 3 via two paths having a length difference of $\lambda/2$ (180° phase difference), resulting in complete cancellation (destructive interference).
- Phase Synchronization: In the third diagram, we observe that at 5 GHz the phases of the voltages at Port 2 and Port 4 coincide (around -90°). Consequently, the two output signals are perfectly in phase.

2. "Good Operation" Area (Bandwidth): A hybrid ring does not operate ideally over an infinite range. As a "good operation" area we define the spectrum in which satisfactory matching ($|S11| \leq -15$ dB) is maintained and the isolation remains high (the power at Port 3 remains near zero).

- Observing the first graph, the $|S11|$ curve remains below -15 dB in the range approximately from 4.2 GHz to 5.8 GHz.
- In this range, the powers at ports 2 and 4 remain relatively balanced (small deviation from 3dB).
- Therefore, the range of good operation of the circuit is about 4.5 GHz−5.5 GHz, offering a relative fractional bandwidth of the order of 20%, which is also the typical theoretical limit for Rat-Race couplers.

1.4. Microwave amplifier (conjugate matching)

A two-stage microwave amplifier operating at 2.5 GHz is designed to operate with maximum possible gain. Suppose we are required to design the circuit connecting the output of transistor A to the input of transistor B, using simple 50 Ω microstrip lines (no isolators possible). The circuit will consist of a simple line and a parallel open-circuited stub, which will have a relatively short length (less than λ/4). The transistor manufacturer gives the values of the reflection coefficient (with reference to 50Ω) at the output of transistor A, $S_{22}^A = 0.72\angle 140°$ and at the input of transistor B, $S_{11}^B = 0.44\angle 162°$.

(a) Design the circuit that leads to the shortest possible length of the connection line: calculate (in λ), the length of the connection line, the length of the stub and its position (either at the output of A or at the input of B).

Solution using the tool: https://onlinesmithchart.com
Solution link:
https://onlinesmithchart.com/?zo=1&circuit=blackBox_0.1837_-0.353_0__transmissionLine_0.0903_%CE%BB_0_1_1__stub_0.179_%CE%BB_0_1_1&zMarkers=0.1837_-0.353__0.397_0.134&frequency=2.5&frequencyUnit=GHz

To achieve the maximum possible gain in the two-stage amplifier, the implementation of a conjugate matching circuit at the interface between the output of transistor A and the input of transistor B is required.

1. Parameter and Target Definition

The values of the reflection coefficients (with reference to 50Ω) are:
Transistor A Output: $S_{22}^A = 0.72\angle 140°$
Transistor B Input (Load): $S_{11}^B = 0.44\angle 162°$

The goal of matching is to transform the input impedance of transistor B so that the reflectivity Γ_{in} is the complex conjugate of the output of A:
$\Gamma_{in} = (S_{22}^A)^* = 0.72\angle -140°$

2. Mathematical Calculation of Resistances

The conversion from the reflection coefficient Γ to the normalized complex impedance z is done via the relationship $z = \dfrac{1+\Gamma}{1-\Gamma}$. Based on exact calculations:

For Transistor B (Start MK0):

$$z_L = \frac{1 + 0.44\angle 162^\circ}{1 - 0.44\angle 162^\circ} \approx 0.397 + j0.134$$

For Transistor A (Target MK1):

$$z_{target} = \frac{1 + 0.72\angle - 140^\circ}{1 - 0.72\angle - 140^\circ} \approx 0.183 - j0.353$$

3. Graphical Solution on Smith Chart

The simulation in the tool https://onlinesmithchart.com was carried out with the following steps:

1. Series Line Rotation: Starting from Transistor A (MK1), we perform a clockwise rotation on the constant VSWR circle. The rotation stops at the first intersection with the constant conductance circle passing through the target (MK0).
The minimum length of the connection line 50Ω was calculated to be: d=0.053λ

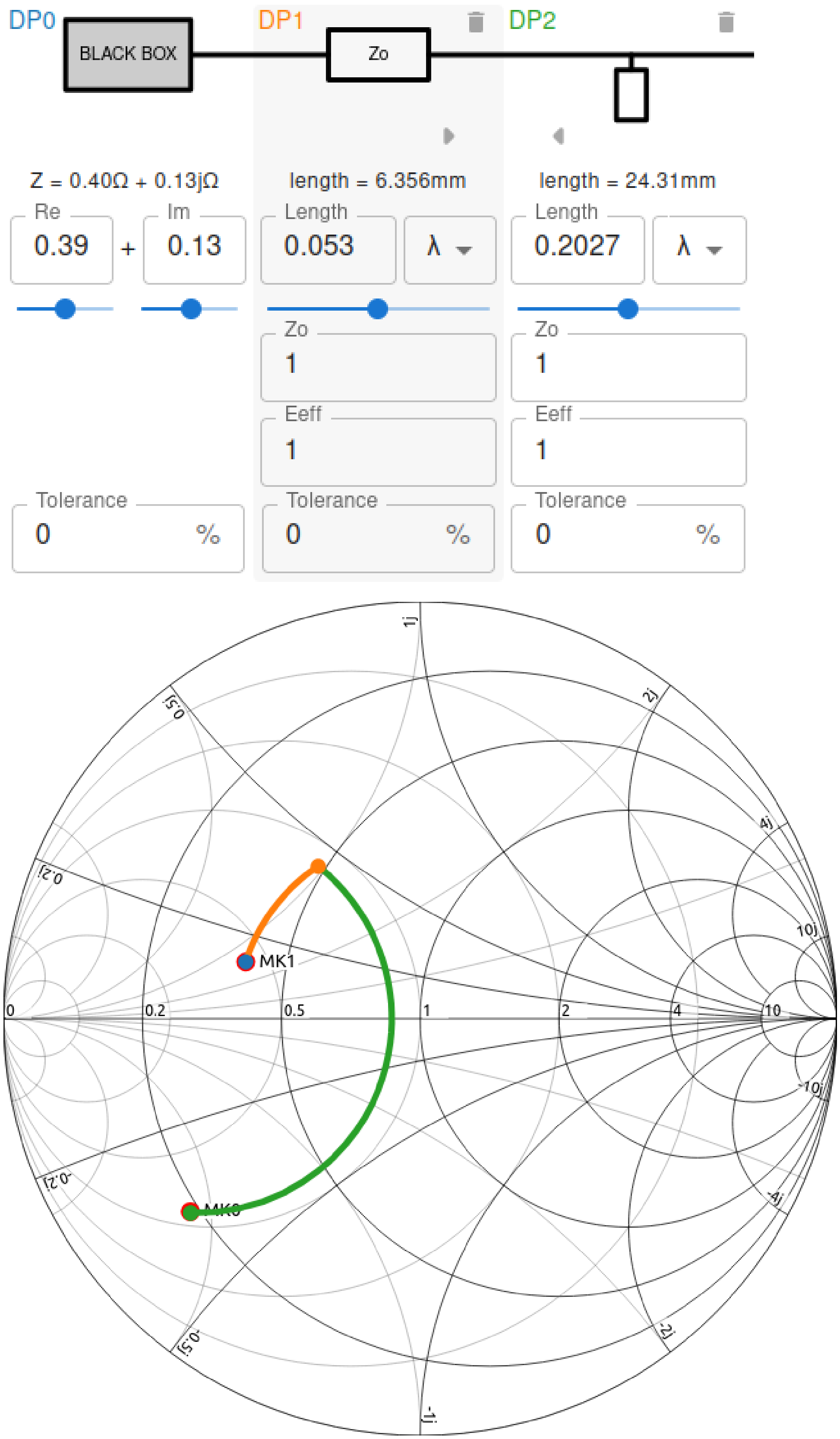
2. Shunt Stub Addition: At this point, a parallel open-circuited stub is added. The stub's susceptance moves the point along the constant conductance circle until final coincidence with point MK1.
The stub length was calculated as: l=0.2027λ

4. Documentation of Minimum Length

This solution is optimal (shortest) as the first point of intersection during clockwise rotation was chosen. Additionally, the stub length l=0.2027λ and the line length d=0.053λ are less than the limit $\lambda/4$ (0.25λ), satisfying the constraint for small physical size and minimization of losses.

Impedance markers ◉ Rectangular ○ Polar

Name	Real	Imaginary	Add
	0	0	⊕
MK0	0.1837	-0.353	
MK1	0.397	0.134	

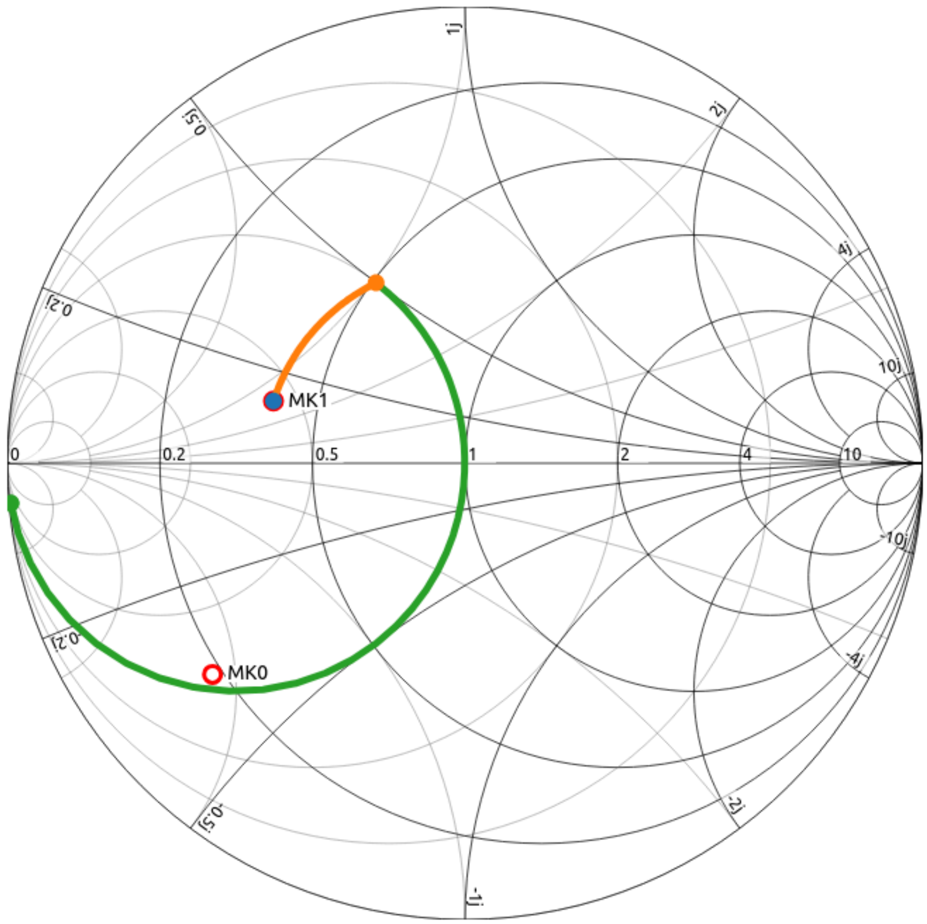

DP0
BLACK BOX
DP1
Zo
DP2
Z = 0.40Ω + 0.13jΩ
length = 6.356mm
length = 24.31mm
Re
0.39
+
Im
0.13
Length
0.053
λ
Length
0.2027
λ
Zo
1
Zo
1
Eeff
1
Eeff
1
Tolerance
0
%
Tolerance
0
%
Tolerance
0
%
MK1
MK0
0
0.2
0.5
1
2
4
10
1j
2j
4j
10j
0.5j
0.2j
-0.2j
-0.5j
-1j
-2j
-4j
-10j

(b) If at 3 GHz, the characteristics of the transistors do not change, find by how many dB the power delivered to the second transistor decreases (and therefore the gain of the amplifier decreases by the same amount).

Solution using the tool: https://onlinesmithchart.com
Solution link:
https://onlinesmithchart.com/?zo=1&circuit=blackBox_0.397_0.134_0__transmissionLine_6.356_mm_0_1_1__stub_24.31_mm_0_1_1&zMarkers=0.1837_-0.353__0.397_0.134&frequency=3&frequencyUnit=GHz

With the change in frequency to 3 GHz, the physical lengths of the lines remain constant, but their electrical lengths increase. As seen graphically on the Smith chart, the matching path "misses" significantly and no longer ends at the target.

For the tool I use, I changed the λ to mm that it had calculated for 2.5GHz (6.356mm and 24.31mm respectively), and changed 2.5GHz to 3GHz. So I also got the results:



Specifically, the simulation shows that the reflectivity at the output of the matching circuit (the end point of the green line) is:
$0.996\angle -175.0^{\circ}$

Reflection Coefficient	-0.992 - 0.088j	0.996 ∠ -175.0°	Q Factor	22.61

The actual load that the circuit must drive is transistor B (point MK0 on the chart), with given reflectivity:
$S_{22}^{A} = 0.72\angle 140^{\circ} = \Gamma_{load}$

The reduction in delivered power is calculated via the Mismatch Factor between these two points:

$$M = \frac{(1-|\Gamma_{out}|^2)(1-|\Gamma_{load}|^2)}{|1-\Gamma_{out}\cdot\Gamma_{load}|^2}$$

Calculating the individual terms with the new data:
Numerator:
$(1-0.996^2)(1-0.72^2) = (0.00798)(0.4816) \approx 0.00384$
Product:
$\Gamma_{out}\cdot\Gamma_{load} = (0.996\times 0.72)\angle(-175.0^{\circ}+140^{\circ}) = 0.7171\angle -35.0^{\circ}$

Converting to rectangular form:
$0.7171(\cos(-35.0°) + j\sin(-35.0°)) = 0.5874 - j0.4113$

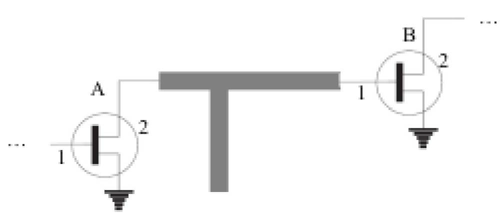
Denominator:
$|1 - (0.5874 - j0.4113)|^2 = |0.4126 + j0.4113|^2 = 0.4126^2 + 0.4113^2 \approx 0.3394$

Consequently:
$M = \dfrac{0.00384}{0.3394} \approx 0.0113$

The gain reduction in dB is:
$\Delta P = 10\log_{10}(0.0113) \approx -19.5 \text{ dB}$

Conclusion: The power delivered decreases dramatically by 19.5 dB. This huge loss is due to the fact that the magnitude of the output reflectivity (0.996) is practically on the periphery of the Smith chart, causing almost total power reflection.

(c) Design the same circuit, if the length of the connection line must be (for construction reasons) necessarily λ/4 (give an approximate solution).



Solution using the tool: https://onlinesmithchart.com
Solution link:
https://onlinesmithchart.com/?zo=1&circuit=blackBox_0.397_0.134_0__transmissionLine_0.25_%CE%BB_0_1_1__stub_0.179_%CE%BB_0_1_1&zMarkers=0.1837_-0.353__0.397_0.134&frequency=2.5&frequencyUnit=GHz

In the case where the connection line has a fixed length $d = 0.25\lambda$ (due to construction constraints), the load point z_L (Transistor B) is transformed through a full 180° rotation on the Smith chart. The new normalized admittance at the line input equals the normalized complex impedance of the load:
$y_{line} = z_L \approx 0.397 + j0.134$

The ideal conjugate target (Transistor A) has normalized admittance:

$y_{target} \approx 1.16 + j2.22$

As we observe, the λ/4 line places us on the constant conductance circle $g \approx 0.40$. Since the target is on the circle $g = 1.16$ and the parallel stub cannot change the real part of the admittance, absolute matching becomes impossible.

Best Approximate Solution:
We use an open-circuited stub (open stub) to fully equalize the imaginary part (susceptance). The required susceptance that the stub must introduce is:
$b_{stub} = b_{target} - b_{line} = 2.22 - 0.134 \approx 2.086$

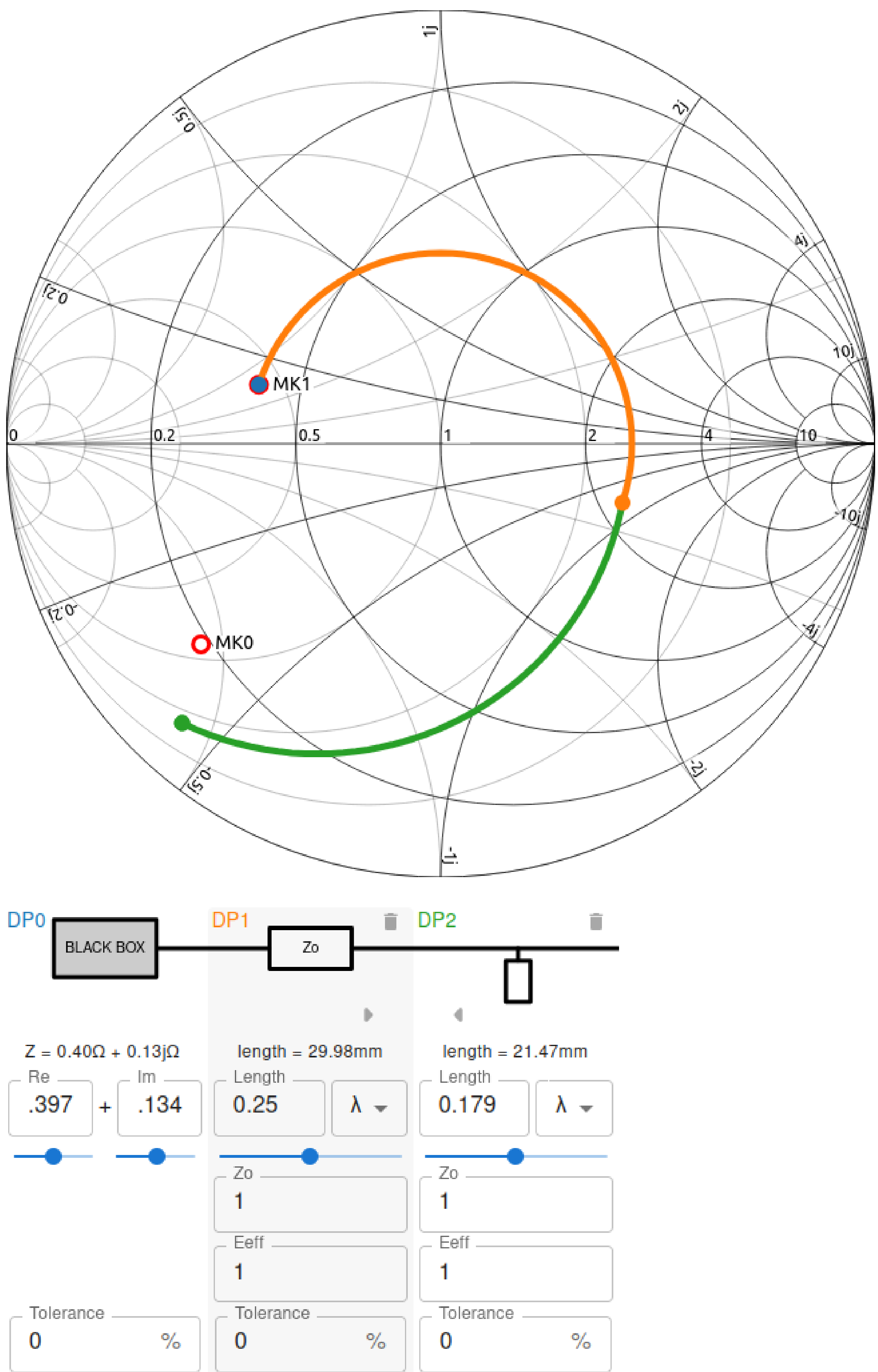
From the susceptance relation of the open-circuited stub we calculate the required length l:

$$jb_{stub} = j\tan(\beta l) \implies 2.086 = \tan\left(\frac{2\pi l}{\lambda}\right)$$

$$l = \frac{\arctan(2.086)}{360^\circ}\lambda \approx 0.179\lambda$$

Design Summary:
The approximate circuit consists of the fixed length connection line $d = 0.25\lambda$ and an open-circuited parallel stub of length $l = 0.179\lambda$.

Diagram Commentary:
In the attached Smith chart, the orange path (0.25λ) transfers the load to the opposite side of the map. The green path (stub) moves along the constant conductance circle $g = 0.4$ and stops at the point where the imaginary part coincides with that of target MK1. The remaining visual distance between the end point of the green line and point MK1 represents the unavoidable matching error of the approximate solution.


1j
2j
0.5j
0.2j
4j
10j
0
0.2
0.5
1
2
4
10
MK1
MK0
-0.2j
-0.5j
-1j
-2j
-4j
-10j
DP0
BLACK BOX
DP1
Zo
DP2
Z = 0.40Ω + 0.13jΩ
Re
.397
+
Im
.134
length = 29.98mm
Length
0.25
λ
length = 21.47mm
Length
0.179
λ
Zo
1
Zo
1
Eeff
1
Eeff
1
Tolerance
0
%
Tolerance
0
%
Tolerance
0
%